Parametric Study Exploring Debonding Failure in Epoxy Tied Pre-Tensioned Natural Stone Structures at Elevated Temperatures

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I. ABSTRACT

This research explores a novel system of epoxy tied pretensioned stone developed by Sebastian and Webb in a pilot study published in 2021. The introduction will address the wider context of this system - contemporary issues in the construction industry relating to climate, the potential of natural stone to act as an environmentally alternative to concrete, the underlying principles of prestressing. The literature review begins with an account of the development of prestressing in stone and the research conducted by Sebastian and Webb. Observations from a preliminary fire test conducted by Sebastian and Webb are discussed and act to inform the scope of investigation. The thermo-mechanical performance of stone and epoxy resin are first explored, followed by an evaluation of the models and methods utilised in the field of structural fire engineering and the nature of thermal stressing. The analysis continues to give a comparison of the relative performance of four common building stones: Limestone, Granite, Marble and Sandstone with regards to

debonding failure using a code built in MATLAB that combines numerical FEA techniques for calculating the evolution of temperature over time and space, analytical expressions for converting those temperatures into stresses and temperature dependant functions to capture the evolution of material properties as they heat. Results show a clear ranking in terms of minimum cover requirement and corresponds well tpo the qualitative understanding of the stones developed in the literature review – however the extent to which the process of cracking can be understood from the results produced by the linear elastic equations is limited.

1. Introduction

1.1. The Construction Industry and The Climate Imperative

Buildings account for approximately a third of global greenhouse emissions - within the EU this figure is even higher with the building sector accounting for half of all greenhouse emissions[1]. While the running of buildings is the biggest contributor the construction materials themselves still play a significant role. Subsequently, the decarbonisation of construction is critical if the goal of UN sustainable development target of net-zero carbon emissions is to be reached by 2050. [2]

By virtue of its versatility and durability, Concrete is, besides water, the worlds most used construction material. The global impact of Portland cement alone is estimated to be 7% of total greenhouse gas emissions[3]. Being such a high carbon material, it should have alternatives.

1.2. Natural Stone as An Alternative to Concrete

Natural stone is an ancient building material and many of the world's oldest extant structures are natural stone constructions. The global stock of natural stone heritage structures ranges from primitive post and lintel constructions such as Stone Henge (UK) to the Pyramids of Giza (Egypt), the rock-cut architecture of Petra (Jordan) or the Inca citadel of Machu Picchu (Peru). This is testimony to natural stones excellent durability[4]. This durability, along with high thermal mass and compressive strength puts natural stone in contention with concrete for many applications. These qualities, along with the advent of climate change awareness, has created a renewed interest in stone as a building material that could offer an alternative to concrete.

While natural stone is not a categorically sustainable material, it is estimated that at the current rate of extraction there is a 850-million-year reserve – furthermore formation of new stone in the earth's crust probably exceeds this rate[5]. Stone has a very low embodied energy and embodied carbon when compared with concrete[6] - stone manufacturer Polycor claims that their limestone has 830kg less embodied CO2 than concrete per ton, as well as a no synthetic chemicals or ingredients and simplified supply chain meaning fewer vehicle movements, less mess, less noise, leas pollution and less waste[7]. Furthermore, Stone is also widely considered to be an attractive material meaning construction programmes can be reduced by eliminating the need for post applied finishes. When evaluated wholistically it has many advantages and can be considered an environmentally responsible material – as transport is the main contributing factor to its embodied carbon this is especially true when the stone is sourced locally[5].

Stone is also a naturally occurring anisotropic material characterised by the presence of clear bedding planes meaning mechanical properties will differ in each of the three rift/bedding planes. Due to this high natural variability, the testing of samples taken from the specific blocks of stone to be used in the construction is necessary [6] – these factors make engineering with stone a more involved process compared to concrete. Another difference compared to concrete is that material strength is usually governed by the presence of cracks, planes of weakness, crystalline veins, or inclusions such as fossils. Furthermore, being a high-stiffness low-ductility material, stone is susceptible to brittle failure and crack propagation. It must therefore be treated with a greater degree caution [6].

1.3. Natural Stone and Prestressing

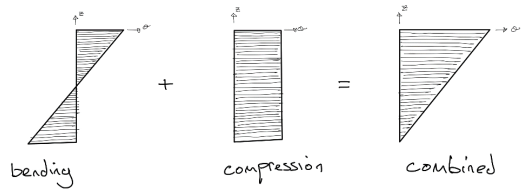

Figure 1: A Simple schematic outlining the underlying principle of prestressing.

The nature of historic natural stone construction and engineering is elucidated in Jacques Heymans' *The Stone Skeleton*. When considering the three main structural criteria of strength, stiffness, and stability, it is the third major criterion that is relevant for masonry; stability is achieved through compressive forces that arise naturally form the structures self-weight and are equally resisted by the stones' excellent compressive strength. Instability occurs due to the low tensile strength of the stone itself as well as the negligible tensile strengths of the mortar joints between the stone - when tensile forces arise cracks form in the masonry that in essence act like hinges creating a mechanism for collapse[8]. Therefore, while traditional masonry building has managed to achieve incredible stability it has been largely constrained to compression-dominated forms - load bearing walls, arches, vaults, domes. These factors have inhibited the use of masonry where flexural stresses may arise. Contemporary architecture demands tall structures that can resist high wind loads and open-plan spaces with the ability to adapt and reconfigure internal arrangements – to realise these structures efficiently flexural strength is necessary.

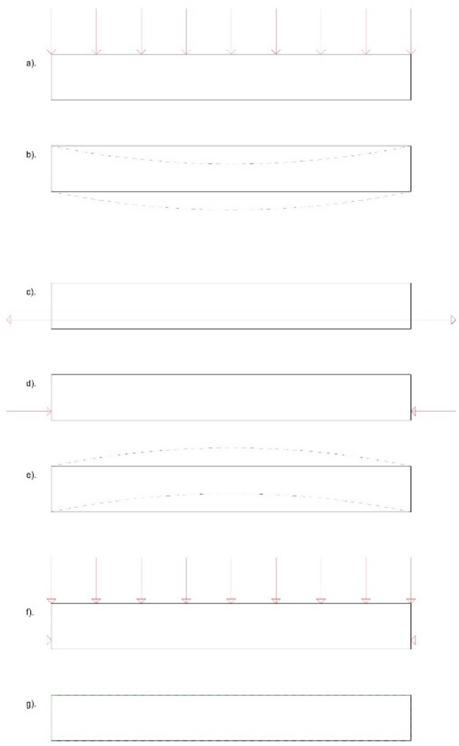

Figure 2: Principles of pre-cambering. a) Beam under applied load. b) Deflection under applied load. c) Prestressed cable force. d) Beam under resultant prestressed force. e) Deflection under prestressed force (pre-cambering). f) Beam under applied and prestress. g) Deflection under applied and prestress force. [Authors Own]

The underlying principle of prestressing is to eliminate bending tensile stresses through artificially induced compression - typically achieved with tensioned steel tendons. The above schematic depicts how the two stresses would superpose assuming the elastic limit is not exceeded. Prestressing utilises stones high ratio of compressive to tensile strength to maximum effect and allows for the creation of stone structures with enhanced bending strengths – tensioning also increases the stiffness of stone structures by reducing joint cracking[9]. Prestressing force can be applied concentrically or eccentrically. Concentric stressing will produce a uniform force, while eccentric stressing will produce a moment equal to the product of the prestressing force and distance from the neutral axis. Eccentric stressing can be advantageous; placement of the prestressing force bellows the neutral axis will result in a hogging moment - causing the beam to deflect upwards. This technique, known as pre-cambering, has the advantage of reducing deflection under loading conditions[10].

These techniques have been widely exploited for concrete structures but it's application with natural stone structures has not been developed to the same extent.[9] This can be partly attributed to the lack of guidance in contemporary structural design codes[6]. Where the technique has been used, substantial structural testing has been required to develop an understanding for its safe use. There are two broad categories of prestressing: post-tensioned and pre-tensioned. Initially developed in concrete, both techniques have now been applied to stone. The development of post- and pre-tensioned stone shall be covered in the literature review – including an account of the system piloted by Sebastian and Webb that this work research addresses.

2. Literature Review

2.1. The development of Prestressed Stone

2.1.1. *Post-tensioned Stone*

Post-tensioning in concrete is where the prestress is applied after the concrete has set. When the concrete is cast a duct is left for the prestressing cable to be inserted into; the cable is then secured at either end of the element using jack and thrust plate connectors. These end plates are what then drives compression into the concrete element[11]. With stone the process is largely the same, but the element is made up of stone segments that have been shaped, cut and drilled. Mortar grout facilitates the transfer of loads between segments.

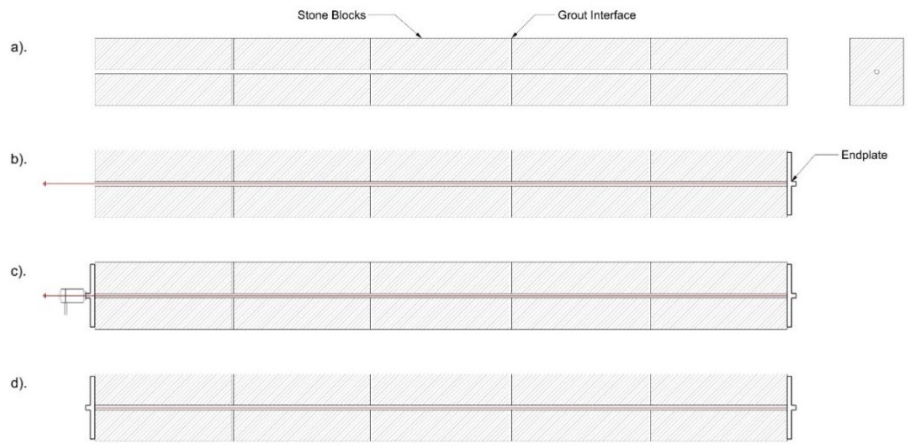

Figure 3: Post-Tensioned Stone Construction Procedure. a) Stone blocks with grout interface and hole bored for cable. b) Cable threaded through hole and affixed to endplate. c) Cable tensioned with jack and plate connection. d) Completed beam, cable tension equilibrated by compression in the stone. [Authors Own]

Engineers Webb Yates have been at the forefront of the development of this system and have authored a series of engineer's guides ("Stone as Structural Material") for ISTRUCTE that act as a principal point of reference in the absence of any formalised codes. One of their earliest demonstration pieces was a 50mm thick bench with a span of 3.2m - in essence a conventional beam it highlighted the potential of pre-tensioned natural stone for flexural applications. But, perhaps the most striking examples of what can be achieved with prestressed stone are the unsupported helical stone staircases they have produced that must resist a complex combination of bending and torsion moments [6], [9], [12], [13].

Figure 4: Post-tensioned Helical Staircase engineered by Webb Yates.

Development of such a complex form required extensive FEA modelling validated with practical tests. A key consideration when designing post-tensioned structures is loss of prestress over time. Loss of prestress comes from three sources: relaxation losses as individual cables lock together, friction between the cable and the hole it runs through, draw-in of the locking wedge. Tension loss due to creeping in post-tensioned concrete was not observed in stone[9].

2.1.2. *Pre-tensioned Stone*

Pre-tensioned stone beams (PSB's) where developed in a study by Sebastian and Webb - prototype specimens were fabricated using epoxy adhesive for longitudinal shear transfer between prestressed steel strands and pre-cut Valange limestone blocks with mortar inter-block joints. Since the strand would be tensioned before the epoxy became structurally active, it is classified as pre-tensioned [10].

This novel system of prestressing stone presents some key advantages over the post-tensioning method discussed previously. The use of a structural adhesive as a strand to stone shear connector would conceal the mechanism of strand-stone force transfer within the stone, allowing end plates to be dispensed of and subsequently leaving the end faces of the stone members clear. The benefit of dispensing of the anchor plates is substantial as exposed steel plates and protruding strands can be problematic if space is limited at the end joint. Post-tensioned stone panels used in the Sagrada Familia had anchor plates sit in recesses machined into the stone block[14]. However, the extent to which anchor blocks can be reduced and concealed is limited as they must be of sufficient size so that no local overstressing of the stone occurs [9]. Furthermore, fabrication could be done onsite or at a factory - making it an agile construction system. One drawback of the pre-tensioning when compared to post-tensioning would be the inability to de-tension the strands, which enables re-use of the stone.

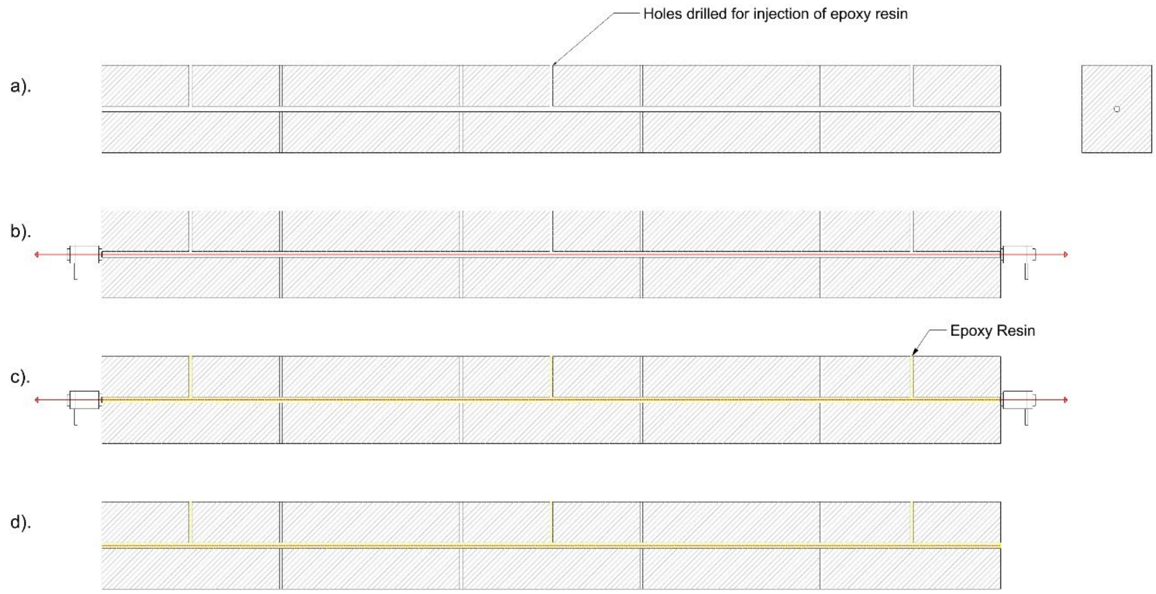

Figure 5: Pre-Tensioned Stone Construction Procedure. a) Stone blocks with mortar interblock joints and holes drilled for pouring of epoxy. b) Steel strand tensioned against stone blocks. c) Epoxy poured while strand is held in tension. d). Tension released once epoxy resin has fully cured. [Authors Own]

Pre-tensioning is typically applied to concrete: strands are tensioned against a casting bed into which concrete is poured; once cured sufficiently the strands are released and their forces are transferred into the bulk of the concrete via a thin concrete zone around the strands. After release there is a transfer zone along which the strand's tensile force, and so the concrete's equilibrating compressive force, increases nonlinearly from zero at the ends to a peak at short distances inwards [10], [11]. Identifying an effective strand to stone shear connection and developing an understanding of the transfer zone mechanics was a principal objective of the study, conducted by Sebastian and Webb.

Epoxy resin was identified as a potentially suitable adhesive as it can be poured wet to flow through predrilled holes and occupy the space between the strand and the stone. On curing the epoxy would become structurally active, adhering to the stone, and developing compression/shear stiffness and strength adequate for the longitudinal shear transfer from strand to block[10].

Pull-out tests were conducted to validate the use of an epoxy resin. A seven-wire smooth surface strand of 15.7 mm diameter, 1860 N/mm2 tensile strength, 150 mm2 section area and a maximum force of 321 kN peak value was fed through a 20mm diameter hole drilled into Valange limestone blocks. A two-part epoxy, with a compressive strength of 70 N/mm2 at 7 days, tensile, flexural and slant shear strengths of 26 N/mm2, 58 N/mm2 and 55 N/mm2 respectively, and a 3.6 kN/mm2 elastic modulus, was then injected with a cartridge via a side hole that intersected with the main 20 mm diameter hole[10].

Four embedment lengths where tested (200, 400, 600, 800 mm). The results show that in all cases it was the strand-adhesive bond that failed as opposed to the stone-adhesive bond. All specimens showed similar values for peak mean bond stress (falling between 4.6 and 5.5 N/mm) and plots of mean bond stress against slip showed consistent behaviour. Furthermore, pull-out load capacity and embedment length showed a good linear correlation with a mean gradient of 245kN/m length of strand. Pull-out capacity should level off at a threshold embedment length – as this threshold was not reached in the experiment suggesting the maximum pull-out capacity of 196kN for 800mm embedment could be increased further. Examination of failed stone specimens showed that the epoxy resin had bonded well to the stone and packed tightly around the steel cables. These results confirmed the internal anchorage potential of epoxy resin and cartridges as an effective method of feeding the epoxy into the stone around the strand [10].

Figure 6: Failed specimen shows bond between epoxy resin and stone [10].

Subsequently, a 3.5m long concentrically prestressed beam was produced to study the transfer zone mechanics. Steel tendons where tensioned against the stone sections with 200kN of force using a jack and endplate. The of epoxy was then applied and given 10 days to cure, at which point anchorage was removed and all force was transferred from strand-to-stone via the epoxy adhesive. Longitudinal strain gauges placed at the midpoint of all four faces at various points along the beam. A plot of mean compressive strain along the length of the beam show that the transfer length was between 600 mm and 800 mm, further confirming the effectiveness of epoxy resin as a shear transfer mechanism [10].

Vibration tests conducted on concentrically and eccentrically prestressed specimens which revealed damping of 3.5% and 3.8% respectively – a level useful for the floors of buildings. A three-point bending test was conducted on an eccentric strand specimen – cracking at the midspan mortar joint was delayed to 40% of peak load showing a clear benefit of pre-tensioning. The beam ultimately failed due to compression loads at the midspan – spalling was followed by asymmetric crack that propagated from the midspan. Further testing is needed to confirm the consistency of this failure mode [10].

Figure 8: Photos of fire test.

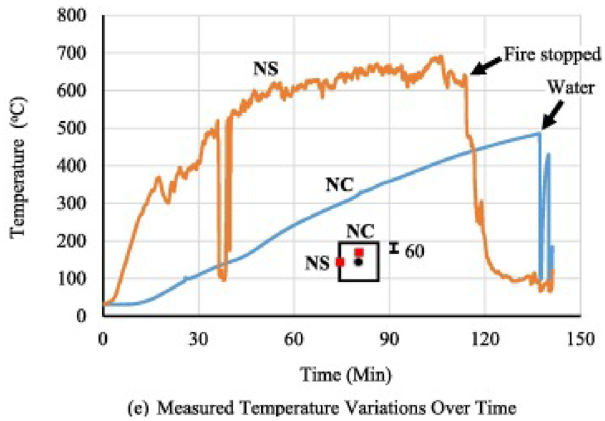

Figure 8: Recorded near surface (NS) and near core (NC) temperatures.

A final, and destructive, fire test was carried out on the concentrically stressed specimen. Supported near its ends the beam was exposed to fire from a drum fed by wood directly below its midspan. The test lasted nearly 2.5 hours, temperature variation was measured by two thermocouples: a "near-surface (NS)" thermocouple was placed just under the surface of a vertical face of the block, while a "near-core (NC)" thermocouple was placed about 5 mm above the core longitudinal hole in that block[10].

The following observations where made:

1. No explosive failure was observed.
2. 85 minutes into the test longitudinal cracks were observed on the top and side surfaces of the beam. Soon after a small flame was observed to emanate from the top crack.
3. After the test, it was observed that an approximately 2mm thick layer had peeled from the bottom face of the Valange that had been directly exposed to the flame. When the blocks where subsequently broken open the epoxy at the midspan had blackened and degraded/burnt, epoxy adjacent to this had begun to discolour while epoxy further from the fire remained visually unaltered.
4. There was a substantial lag between NC and NS temperatures.

This test was essentially qualitative in nature but gives a picture of the general response of the system, reveals vulnerabilities, and informs future research.

1. Longitudinal cracks suggest that further exploration should be directed towards understanding how fire induced thermal stresses can lead to cracking that exposes the epoxy adhesive to direct fire.
2. Thermal degradation of the Valange blocks directly exposed to fire was moderate but not severe. Furthermore, as spalling was only observed after the test was concluded it is possible that it was caused by the thermal shock of being abruptly cooled with water rather than fire induced thermal stresses.
3. The substantial lag between the NC and NS temperatures suggests the stone acted as a good insulator. However, NC temperatures still reached a substantial 500 C – further research is needed in defining

the operative temperature of the epoxy and determining the minimum cover required for adequate insulation of the epoxy.

4. Surface temperatures (NS) did not reach those modelled by the ISO fire curve - further research should simulate a building fire as opposed to an arbitrary fire.
5. The fire test was also an unloaded test, limiting its usefulness when drawing conclusions about how it would behave in service.
6. The fire was highly localised as opposed to widespread, also limiting the scope of conclusions that can be made.

The results of Sebastian and Webb's pilot study confirm the suitability of epoxy as the strand-stone adhesive but highlight a gap in knowledge regarding the fire safety of the novel strand-epoxy-stone component. Subsequently, literature regarding thermal effects on stone, thermal effects on epoxy, the principles of thermal stresses and the state of structural fire engineering should be reviewed. Furthermore, the literature review should address the range of stones used in buildings so that their relative fire performance may be evaluated.

2.2. Thermal Effects on Stone

Rocks can be considered as a dual space composed of solid material (i.e. minerals and binding matrix) and void space [15]. The geometry and density of these cracks and pores greatly determines the physical parameters of the rock. Temperature variation in rock introduces new crack and microcrack development and is therefore one of the primary factors in determining the integrity and physical properties of rock [16]. As a result, although natural stones are non-combustible materials, the effect of fire and heat can cause irreversible changes in their structure and mechanical behaviour, which influences strength/stiffness, and these changes can threaten the stability of the entire structure [17].

Despite this, until the end of the 20th century most investigation into the fire resistance of construction materials had been focused on wood, steel and concrete [17]. Ensuing research on stone was concerned with the conservation of historic monuments and sought to establish, in the case of a fire, on which basis stone can be restored without replacement and when replacement is necessary [18].

Figure 9: Fires at Windsor Castle (1992) and Notre Dame (2019). Research into fire damage to natural stonework has been accelerated by some fire incidents that threatened historic monuments [17]

Early research was qualitative in nature and mainly focussed on morphological changes taking place on stone surfaces such as cracking, scaling or analyses of colour changes. A review of such research conducted by BRE concluded that at high temperatures (600-800 C) most building stones are seriously damaged. At lower temperatures (200-300 C) damage is restricted to colour changes[18]. Due to stones low thermal diffusivity, high temperature gradients occur – thermal shock upon cooling therefor presents the greatest danger as it can completely disintegrate the stone [19]. The review also concluded that the behaviour of building stones varies greatly between major stone types: Limestone, Marble, Sandstone, Granite [18]. The behaviour of each shall be investigated in further detail in the following sections.

More recent research increasingly gives a more quantitative description of the thermal effects on stone – measuring physical and mechanical properties across a range of temperatures. However, the structural application of stone and the effect of building fires is not the primary motivation for much research into thermal effects on stone. Therefore, few studies both cover the full range of fire temperatures (up to 1000 C), test all the mechanical properties needed for structural analysis (Youngs Modulus, Poisson's Ratio, Compressive and Tensile Strength), as well as the temperature dependant thermal properties required for heat analysis (thermal heat capacity, thermal conductivity, mass density) [20]. A 2018 review of assessment techniques for studying the effects of fire on stone materials evaluated studies from the preceding 30 years – of the 39 studies on fire only 10 tested up to 1000 C, of these only three measured the Young's Modulus, three measured tensile strength and five compressive strength – no study measured all three parameters[21]. At lower temperatures more complete sets of data can be found – for sandstone, limestone, and marble up to 800 C [22]; for sandstone and limestone up to 400 C [23].

Data is highly fragmented and must be compiled from multiple sources. Although almost all studies are conducted in laboratory conditions – allowing for the replication of standardised tests – researchers use diverse experimental set ups, sample dimensions/geometries, heating rates/thermal gradients, exposure times and cooling conditions. Experimental variation is not trivial, for example heating rates ranged from 1 to 30 C/min – a more rapid heating rate will induce a steeper thermal gradient in the stone leading to more significant physio-mechanical modifications. This variation, compounded with the natural variation of the stone itself, leads to complication when comparing/compiling data and "makes it impossible to determine how stone each stone will evolve with heating" with accuracy [21].

Application of data, regarding the thermal response of stone, to the case of a building fire is also limited by the nature of laboratory setups where the stone is typically heated via convection – in a real fire the stone would be heated by radiation. Furthermore, to the authors knowledge, there are no studies on the effects of fire on aged stones – it is known that interaction between stone and environmental agents (e.g., rain and atmospheric pollutants) can have a strong effect on stone causing physical-mechanical and mineralogical modifications [21]. Finally, no studies that address the anisotropic nature of stone (measuring thermal characteristics for each orientation to the bedding plan) have been found by the author.

From the literature reviewed so far it is clear there is a deficit of data regarding the thermal response of stone. Establishment of a test method/standard for determining the resistance of stones to elevated temperatures, and a database for collation of results, is essential for the development of natural stone engineering. In the following sections specific behaviour of marble/limestone and granite/sandstone shall be discussed.

2.2.1. *Marble and Limestone*

Highly coveted for its aesthetic qualities, Marble is a stone that has frequently been used in construction – for both decorative and structural purposes[24]. Limestone has likewise been widely exploited as a construction material and was the stone used by Sebastian and Webb to fabricate prototypical pretensioned beams [10]. Both marble and limestone are carbonate rocks but while limestone is sedimentary limestone is metamorphic.

Being carbonate rocks the primary mechanism for thermal degradation is the same – microcracking occurs due to internal stress concentrations, which are caused by the anisotropic expansion of calcite crystals/grains [16]. The severity of this action depends on grain size, porosity and structural characteristics – marble is in general more vulnerable due to its higher density and inability to absorb internal stresses. In extreme cases it can be so friable that it crumbles to powder [16], [19], [25]. Microscopic analysis of limestone and marble samples show that when heated up to 150 C the dilation of calcite grains causes rock compaction, that can increase peak strengths [16]. Similarly, one study showed that when heated to 240 C uniaxial compressive strength increased with temperature [26]. When heated further, to 800 C, the stones structure becomes damaged and the rock can be observed to crack, fragmentize, spall, and disperse generally [27]. At these temperatures the effects of the calcite expansion are further compounded by the decomposition of calcium carbonate which only occurs between 700 C and 800 C [19] and possibly also by an endothermic reaction between calcine and water[27].

A study of Turkish marble and limestone shows that residual tensile and compressive strengths were reduced substantially as temperature is increased beyond 400 degrees [24]. Data for young's modulus is scattered for marble but exists for limestone. Regarding thermal properties data is generally available – there are gaps for specific heat capacity of limestone between 300-500 C and very little data for marble. While the thermal expansion of limestone has been studied across a good range of temperatures there is a lack of information about marble beyond 700 C. Further research is required on how poisons ratio changes with temperature for both stones[28].

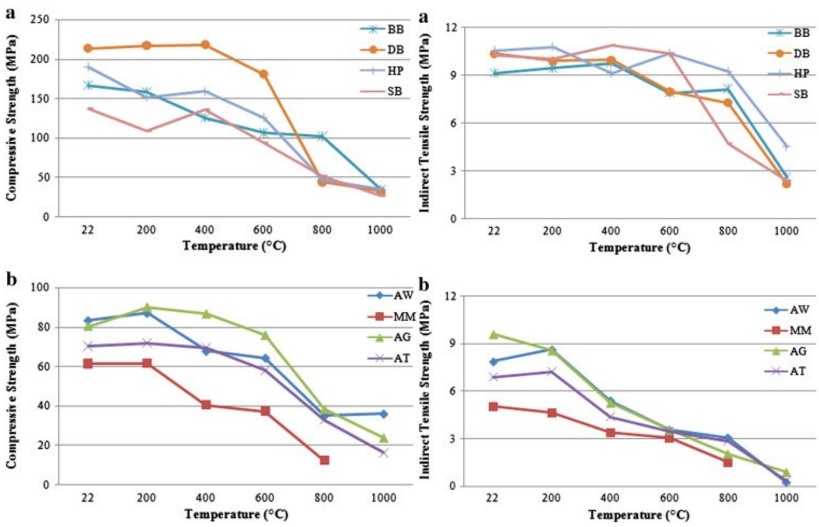

Figure 10: Temperature effect on Tensile and Compressive Strengths. a) Limestones, b) Marbles. [24].

2.2.2. *Granite and Sandstone*

Behaviour of granite and sandstone is substantially different to the aforementioned carbonate rocks. When heated to high temperatures granite and sandstone are vulnerable due to a quartz component that undergoes a reversible change in crystal structure at 573C - putting the stone at risk of cracking or shattering [25], [29]. This effect is more severe for granites, due to their high density and lack of void space they are unable to absorb the internal stresses that arise with the thermal expansion of mineral grains. The dehydroxilation of phyllosilicates and clay minerals (possible from 200 °C on), is another mineralogical change that governs these stones behaviour when heated[28].

When exposed to heat the main mineral composition of granite does not change with increasing temperatures and thermal expansion from 20C to 250C has been shown to be fully reversible [25]. Further studies have shown a mixed performance at moderate temperature; triaxial compression tests on granite specimens heated up to 500 C showed that once cooled mechanical strengths where slightly improved - this was attributed to the closing of microcracks during heating. However, this thermal treatment had a negative effect on stiffness as young's modulus and poisons ratio where reduced [30]. A study of construction stones quarried outside Madrid showed that low temperature cyclical loading could have an adverse effect on stiffness and surface hardness – as exposure to 42 thermal cycles (105-20C) led to the propagation of microcracks [31].

A data compilation of previous studies gives the most comprehensive description of the granites thermo-mechanical behaviour up to 1000 C. It shows that some material parameters, such as coefficient of thermal expansion, change abruptly with the a-b transition of quartz [32]. For sandstone more research is needed for the thermal conductivity and specific heat capacity beyond 400 C – but there is food information regarding thermal expansion[28].

Research on the residual tensile strength of sandstones and granites are scarce and more research is needed in this area. Research on the residual compression strengths show that granite decrease significantly from 400C while in sandstone they tend to decrease from 500C onwards[28]. However, a study of sandstones up to 900C showed that different sandstones exhibit widely varying compressive strength when heated depending on mineral composition - carbonate sandstones showed the most significant reduction in compressive strength [33]. Data on the young's modulus of sandstone and granite is scattered beyond 400 C and needs more research, while for data for poisons ratio is scarce for all temperatures[28].

2.3. Effect of Fire and Temperature on Epoxy Structural Adhesives

In civil engineering epoxy resins are used in composite structures to bond various materials together such as Carbon Fibre Reinforced Polymer (CFRP) sheets, concrete, metals, and timbers. Epoxy resins is typically mixed and applied on site only developing sufficient strength and stiffness after a certain curing time[34].

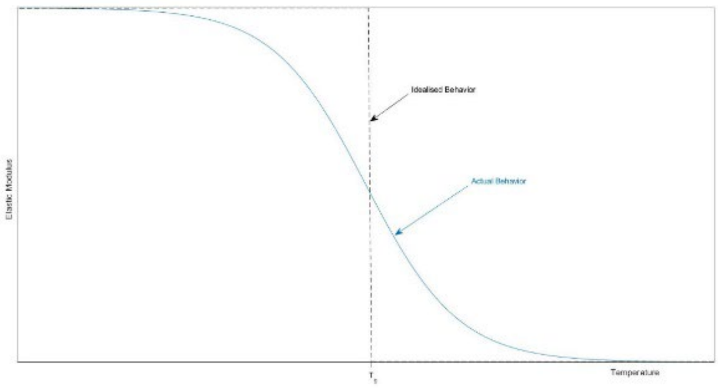

Figure 11: Schematic of Actual and Idealised Glass Transition Behaviour. [Authors Own]

Cured epoxy resins will lose their stiffness and strength again when being subjected to an increased temperature as they undergo 'glass transition' – this is a continuous process that takes place over a temperature range of 10-20 C in which the resin transitions from a solid "glassy" state to a viscous "rubbery" state. The gradual transition between solid and viscose state is idealised as a binary transition that occurs at the glass transition temperature (T_g). This metric is typically supplied by producers and demarcates the maximum operational temperature of the epoxy resin [34] – no such metric was supplied for the resin used by Sebastian and Webb in their pilot study.

Figures given for the glass transition temperature of thermosetting structural adhesives range from 45 to 120 degrees [35]– however experimental studies, focusing specifically on epoxy, tend to give figures on the lower side of this spectrum. Multiple studies have shown that at temperatures above 55-65 C the shear bond strength of structural epoxy resins is reduced by approximately 90% [36]–[38]. This phenomenal loss of as the adhesive reaches glass transition has been commented on in many studies - where a structural connection has been exposed to elevated temperatures the adhesive molecules have oft been found to be the weakest link, exhibiting a steep degradation, and thereby putting the structure at risk [39], [40]. Pull out tests of adhesive coated steel rods embedded in concrete specimens under fire conditions show that bond strength is a function of temperature, and that compared to uncoated bars at the same elevated temperatures, the rate of slip is increased, and residual bond strength decreased[41]. Pinoteau et al. observed that under fire conditions water migrates towards the inner core of the concrete and creates a water layer over the adhesive surface leading to failure at a lower bond strength[42].

Table 1: Mechanical and Physical Data according to the supplier's data sheet.[34]

Resin	Glass Transition Temperature [˚C]	Density [g/cm^3]	Youngs Modulus [MPa]	Comp. Strength [MPa]
S&P Resin 220	54.5-62.1	1.7-1.8 at 23˚C	>7100	>70
Sikadur 30	62	1.65 at 23˚C	9600 (Compression), 11200 (Tension), after 14 days cure at 23˚C	85-95 after 7 days at 35˚C
Sikadur 30 LP	45	1.8 at 20˚C	10000 after 14 days at 25˚C	>85 after 7 days˚C

It should be noted that glass transition temperature of epoxy resins depends on several curing and testing parameters and is as such not a defined material property [43]. A study of epoxy resins produced for the construction industry found glass transition occurred in the region of 40-50 degrees C – considerably lower than the values reported by producers; the study noted the lack of information on many technical sheets regarding which curing, testing, and evaluation parameters were followed for the determination of T_g.[34] . These values where corroborated by another study that tested two commercially available epoxy adhesives produced for structural strengthening, which placed T_g ranged between 41.57 – 56.11 C [43].

Higher specimen age, higher ultimate temperature during testing, accelerated curing, as well as a lower heating rate implicate higher glass transition temperatures [34]. Therefore, with the correct curing process it is likely that higher glass transition temperatures are achievable, however, a specific feature of the construction industry is that works are done on site, regardless of environmental conditions – and subsequently curing conditions are hard to control. In pre-fabrication the curing would be done in a factory setting – meaning high T_g values could be achieved with greater consistency.

When heated beyond T_g and exposed to extreme heat structural adhesives may ignite, supporting flame spread and emitting toxic smoke [44]. – as observed by Sebastian and Webb.

From a review of the literature, it is evident that the appropriate temperature limits to ensure adequate residual performance in structural adhesives are not currently known [44]. Furthermore, if adhesives are applied on site, and curing conditions are not controlled, there will an element of uncertainty regarding the glass transition temperature of the resin. Subsequently, it is clear there must be more research into the glass transition of structural epoxy resins – this research should also take into account the conditions in which the epoxy is cures, on-site or in a factory, as this will affect the distribution of glass transition temperatures realised in practice and it is possible that the characteristic value shall differ between the two scenarios.

2.4. Thermally Induced Stresses

Structural response to fire has long been understood in terms of strength loss by thermal degradation, where “strength” and “load” are considered the key factors, this approach is fundamentally no different from ambient behaviour. While thermal degradation is a relevant parameter that must be considered, findings from full-scale fire tests suggest at Cardington UK posits that thermally induced forces and displacements are the critical factors that govern structural response at elevated temperatures.

Heating, unavoidably, leads to thermal strains being induced in a structure. Under an average centroidal temperature rise these strains take the form of thermal expansion to an increased length, while a temperature gradient through the section depth will lead to curvature. The nature of the of the thermal strains is a function of the fire scenario – a fast burning fire that reaches high temperatures quickly and then dies out could produce lower mean temperature with a steeper thermal gradient compared to a slow burning fire that maintains a lower temperature for an extended period and produces a higher mean temperature with a less pronounced gradient.

The response of the structural member to these strains is determined by the restraints placed upon on it. If the member is unrestrained there would be a displacement-dominated response while more rigid restraints would produce a force dominated response. For example, a beam under a thermal gradient will exhibit thermal bowing when its ends are rotationally free, if its ends are rotationally restrained mechanical stresses will oppose thermal strains and the member will experience large bending moments without deflection. Furthermore, thermally induced curvature would produce tension in a member whose ends are rotationally unrestrained but restrained in translation[45].

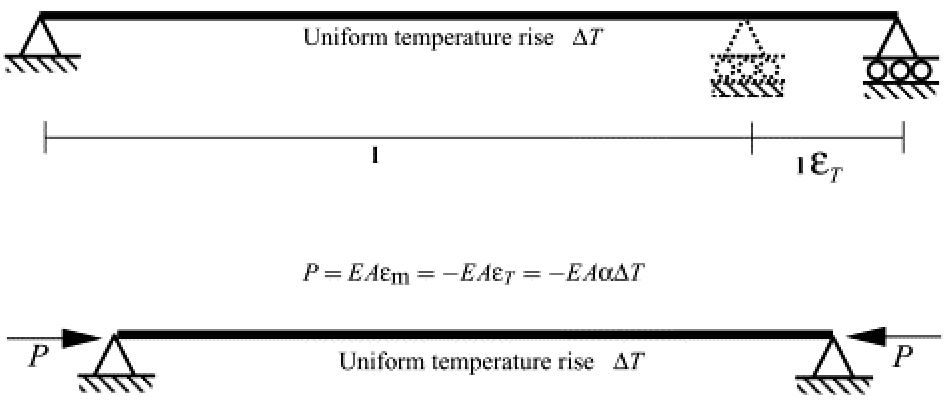

Figure 12: Uniform extension against rigid restraints presents the simplest example of how thermal stresses arise.

Heating of structures therefor leads to a complex mix of mechanical strains, some due to applied loading and some due to restrained thermal expansion, that often exceed yield values and cause plasticisation. The deflection of the structure is dependent on the sum of strains, therefor rigid restraints, while producing smaller deflections, are associated with higher plastic straining and have an adverse effect on the stiffness properties of the material. In a member where thermal strains are free to develop deflections shall be larger by there will be a reduced demand for plastic straining, hence causing less damage to the material. This can produce structural situations that are counterintuitive from a conventional/ambient temperature structural perspective[45].

The above framework is suitable for calculating stresses oriented along the members length. However, even when an element is unrestrained at the supports, as in the fire test conducted by Sebastian and Webb, thermal stresses can still arise and cause failure. This occurs when there is a temperature gradient within the member section causing differing rates of thermal expansion. Parts of the solid that experience greater thermal strains are still restrained by the attached solid – hence mechanical stresses within the section plan develop[46].

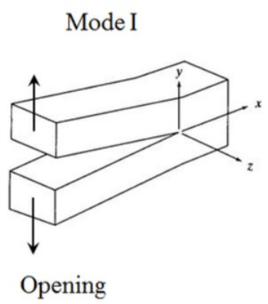

Figure 13: Mode 1 Opening Fracture[47].

The lateral cracking seen in the specimen tested by Sebastian and Webb was of mode 1. And would therefor have been caused by stresses orientated in the section plane (x-y) as opposed to stresses orientated along the member length[46]. Therefor the analysis shall not address the stresses along the member length or the effects of end restraints. There are several analytical expressions for calculating these in simple geometric shapes with simple temperature gradients but for a more complex gradients and shapes numerical techniques are necessary[46].

2.5. Structural Fire Engineering: Current State of The Art

Structural fire engineering (SFE) can be considered as a constituent of the wider multi-discipline practice of Fire Safety Engineering (FSE) which can be defined "the application of scientific and engineering principles to the effects of fire in order to reduce the loss of life and damage to property by quantifying the risks and hazards involved and provide an optimal solution to the application of preventive and protective measures"[48]. SFE is the branch of FSE is specifically concerned with the analysis of structures when exposed to fire hazard and seeks to ensure that the building shall be designed so that, in the event of a fire, its stability will be maintained for a reasonable period [49]. The required structural fire resistance is prescribed by codes is either 30, 60, 90 or 120 minutes depending upon building function, height, and the presence of fire sprinklers [49].

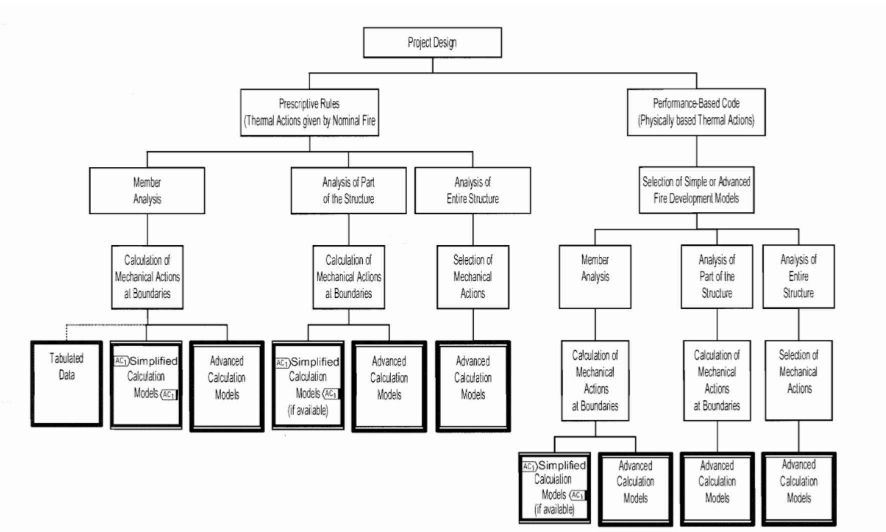

Figure 14. Alternative design procedures, EN 1992-1-2.

Several methods exist for assessing the fire resistance of structures/structural components, ranging in complexity, cost and accuracy the main procedures available are: Standard Fire Tests, Tabulated Data, Simplified calculation methods, Advanced calculation methods, Full-scale fire tests[50].

Standard and full-scale fire tests are the most expensive and resource intensive procedures – putting them beyond the scope of this investigation - but have proved to be greatly insightful and are necessary for the validation of calculations methods. Tabulated data are highly conservative but easy to apply in regular building design, simplified calculation methods are significantly more accurate but are only applicable to a limited range of scenarios. Subsequently, these too are inappropriate for the study of novel construction systems, such as the one under review[50].

Advanced calculation methods in SFE are those in which a time dependant thermal and mechanical analysis is performed to assess the resistance of the whole structure, parts or the structure or simple structural elements Accordingly, the design and assessment of the structure involves three principal steps: [51], [52].

(a) Selection of the "demand" fire scenario/model - typically expressed as gas temperature-time evolution from which boundary conditions are deduced.

(b) Thermal response analysis - based on the acknowledged principles and assumptions of the theory of heat transfer – that considering *the relevant thermal actions* and the *materials temperature dependent thermal properties.*

(c) Mechanical response analysis - based on the acknowledged principles and assumptions of the theory of structural mechanics - that considering the effects of *mechanical properties deterioration with temperature* and *thermally induced stresses and strains*.

2.6. Fire Modelling

Fire is an extremely complex combination of physical phenomena that currently cannot be fully described by means of mathematical models- thus numerous models exist, exhibiting varying levels of simplification [53].

Fire models are implemented in structural analysis with the purpose of determining the evolution in time of the net energy input from the fire to the structure. The net energy input is defined by means of a boundary condition between the structural element and the gas phase – the boundary condition has been considered to be probably the most complex aspect of the SFE process. Fire engineering and structural engineering are distinct domains and with limited interaction between them [54] subsequently the application of inappropriate boundary conditions not uncommon due to incomplete understanding of the underlying physics[55]. Without reliable and appropriate inputs, it is not possible to estimate accurate thermal and mechanical response of structures to fire.

Category			Description	Models
Idealised Models	Uniform	IA	Uniform temperature, all structural components are engulfed in fire, fully developed and no cooling phase described	ISO, ASTM, Hydrocarbon curves
		IB	One steady state maximum temperature over a period of time, ventilation controlled, fully developed fire, uniform temperature	Kawagoe, Thomas, Law, Babrauskas,
		IC	Time-variant temperatures, uniformly distributed temperatures, represents the cooling phase of fire development as well.	Magnusson and Thelandersson, Pettersson, Parametric Curves in Eurocode I, zone models
	Non-uniform	II	Localised fires, fuel controlled, travelling fires	Hasemi's Model, Rein's model, ETFM
Realistic		III	Realistic temperature distributions	Experiments, field models (FSI, AST methods)

Figure 15: Classification of fire models[55]

Numerous models have been proposed by researchers - simplifications and assumptions imbedded in these models mean that they are often limited in scope to a specific fire scenario (compartment fire, travelling fire, localised fire, ect). Furthermore, there is no generalised model that can yield accurate results for all cases and extending a fire model to another scenario is questionable. Fire models have subsequently been classified based on their applicability to a particular fire scenario [55].

Fire models can firstly be classified as idealised or realistic. Idealised models can be further classified into uniform and non-uniform. Uniform models are based on the prediction of average gas temperature of the compartment and do not consider local conditions. This category of can be further divided into three sub-categories:

IA. Uniform, prescriptive time-variant:

These nominal/standard fire curves assume the structure is fully engulfed in flames and do not consider the physics of fire dynamics or heat transfer within the fire compartment. Due to the prescriptive, and highly conservative nature of these models, they are scenario independent. Despite being the earliest and most primitive models, they have been widely used by engineers due to their simplicity, with the standard ISO-fire curve being such an example.

However, these models are non-representative of a realistic fire, and have significant drawbacks: less efficient and cost-effective designs, innovative solutions that could maintaining an acceptable level of fire safety would not be approved by the prescriptive method and it does not develop an understanding of real fire behaviour [38]. Furthermore, they do not include any cooling/decay phase which has been shown to be non-trivial when considering structural response to fire.

IB. Uniform, performance-based time-invariant:

Performance based fire models were developed as in response to the failure of prescriptive models to account for a number of important necessary if the design of structures is to be rational, e.g. quantity and type of combustible material, fuel distribution in the compartment and geometry of the compartment. Built upon FSE principles, they include a simplified description of fire dynamics and heat transfer [40].

This category of performance-based models provides only a single steady state maximum temperature that can be used as a boundary condition for a specified period. Early work by Kawagoe used experimental results to establish the concept of compartment fire and to defined a link between ventilation (geometry) and rate of combustion – from which gas phase temperature is accordingly derived[55].

This would be further developed into Thomas's curve, which relates temperature with opening factor (ventilation characteristic), providing the worst-case time invariant temperature regime for the fire until fuel burnout. His work shows two possible regimes: ventilation-controlled, where the presence of soot and limited supply of oxygen determines the character of the fire, and fuel-controlled, where vents are large, oxygen is plentiful, and it is fuel supply that determines the character of the fire. Law further developed Thomas's work by including the effects of fuel load – firstly arguing that lower fuel loads would result in lower temperatures but more importantly drawing attention to the importance of fire duration when considering fire severity and expressing the time to burnout as a function of burning rate (a function of ventilation) and fuel load. A major limitation of the above models is that they are only applicable in a fully developed ventilation-controlled regime[55].

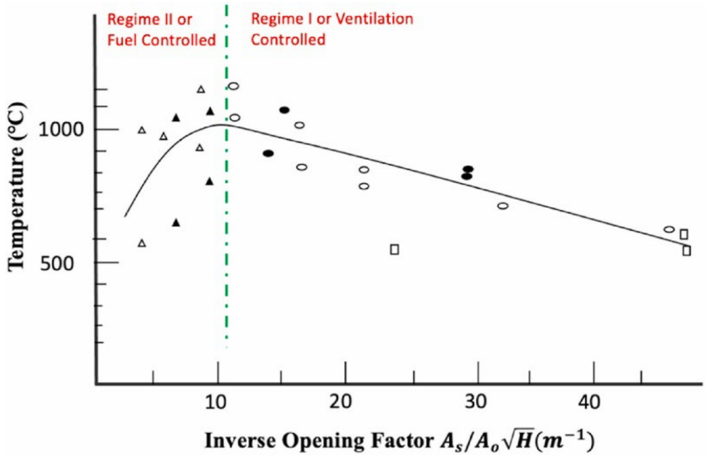

Figure 16: Thomas's curve showing time invariant gas temperatures as a function of opening factor [55]

IC. Uniform, performance-based time-variant.

This category of parametric fire curves – first developed by Magnusson and Thelandersson, then continued by Pettersson - sought to improve upon the preceding models, arguing that the time-invariant expressions only represented the flaming phase and omitted any physical explanation of the cooling/decay phase.

Assuming a fully developed ventilation-controlled regime, uniform temperature in the enclosure, uniform heat transfer coefficients, and one-dimensional heat flow through the bounding surfaces of the structure. The primary goal of their research - a rational description of temperatures during the decay stage of fire – was achieved, taking the form of various parametric curves for different values of opening factor, fire load and structural material. These models, formalised in the parametric curves presented in Eurocode 1, are limited to compartments of 500 m2 or less[55].

II. Non-uniform.

Non uniform models have been developed more recently to address local-fires and travelling fires. The correct modelling of travelling fires - starting locally and spreading across entire floor plates while becoming extinct at locations of origin as the fuel burns out – has become increasingly important as modern office buildings are designed with to have larger continuous open floorplans. A survey of modern buildings found that 92% of the total building volume exceeded the Eurocode limit for its parametric fire curve and experimental data has shown that for larger compartments the assumption of a uniform gas temperature assumption does not hold [56], [57]. Despite this, most modern buildings rely on traditional compartment models[55].

III. Realistic.

In realistic models, fire temperatures are derived from either a field model (CFD) or experimental data. In recent years sophisticated models based on CFD have provided the most realistic representations of fire to date, giving greater spatial and temporal resolution to the thermal boundary conditions of structural surfaces. However, given the complexity and effort involved they are not yet considered to be a practical approach to structural fire resistance design.

2.7. Sensitivity analysis and uncertainty quantification

For highly complex mathematical models (such as the numerical models used for thermal and mechanical response analysis in SFE), relationships between inputs and outputs are not clear, and subsequently the model can be viewed as a black box, i.e. the output is an "opaque" function of its inputs. In such a case, uncertainty that arise from inputs will propagate through the system uncontrolled, limiting confidence in the output of the model. This therefore calls for an evaluation of the confidence in the model - first, a quantification of the uncertainty in any model results (uncertainty analysis); and second, an evaluation of how much each input is contributing to the output uncertainty (sensitivity analysis).

The SFE process has traditionally been deterministic in nature. Design inputs, scenarios and performance criteria are selected on the basis of being considered sufficiently conservative by the engineer. This process does not properly evaluate the safety level (or residual risk) associated with a design as the full spectrum of outcomes and their associated probabilities are not interrogated. Instead, it is assumed that an adequate but unquantified level of safety is attained based on engineering judgment. In the case of exceptional structures/complex buildings (for example those with atypical consequences of failure or those that adopt innovative materials) the simple extrapolation of the professional's experience to (potentially) incompatible scenarios is erroneous – the safety foundation of traditional SFE can only be justified when there have been sufficient real fire events to observe, this is not the case with exceptional structures and subsequently there is a need to explicitly evaluate residual risk [51].

Although probabilistic risk assessment methodologies have gained traction in SFE, the combination of probabilistic methods and advanced numerical simulations has limited progress. Crude methods, such as the Monte Carlo method, that can evaluate uncertainty without bias (without presupposing a distribution of outputs) rely upon repeated random sampling to obtain a distribution of outcomes. The simplicity of such methods makes them accessible and reliable – therefore in the literature they have become the most common approach used to evaluate uncertainties in SFE, used for reinforced concrete columns [58], pre-stressed concrete beams[59], reinforced concrete slabs [60], or steel beams [61].

However, the number of model realisations required combined with the computational time required for high fidelity FEA simulations makes them inappropriate for use practice. The number of simulations needed to accurately predict the failure probability increases as the magnitude of the failure probability diminishes. Using established techniques of Latin Hyper Cube sampling and Monte Carlo methods, a failure probability of 0.01 can be calculated with 20% error using 10,000 samples. While not possible in practice such exercises are still valid in research due to the wealth of data yielded, that may lead to a better understanding of the factors affecting structural fire resistance.

Designers have looked to reduce the number of required realisations by assuming a distribution. However, in absence of a scientific basis this practice may be inappropriate, resulting in a biased assessment of structural performance. Van Coile et al. [62] , for example, found a traditional lognormal approximation to be inappropriate for simply supported one-way concrete slabs exposed to the ISO 834 standard fire. Research has subsequently moved towards developing more sophisticated tools for generating an unbiased quantification of uncertainty with a reduced number of model realisations; a wide range of tools, exploiting convolutional neural networks[63], principles of maximum entropy[51], deterministic sampling and fractional factorial design [64], have shown some success but require further research.

3. Aims and Objectives

The principal aim of the analysis shall be to assess how the mechanical integrity of the epoxy resin adhesive might be compromised in a fire scenario. Two failure modes shall be considered: The debonding of the epoxy resin and longitudinal cracking due to thermal expansion.

The literature review suggests that the debonding of the epoxy would be the dominant failure mode under elevated temperatures. This hypothesis is based on the following: glass transition of epoxy occurs at low temperatures (55-65 C), glass transition results in a catastrophic loss of mechanical strength (90% reduction), as the shear connection fails, the steel tendon will release tension, the beam lose compression and the flexural strength of the system will be negligible causing collapse. Referring to the preliminary fire test conducted by Sebastian and Webb one can see that NC temperatures exceed 100 C at 30 minutes, while thermally induced cracking only occurs at 83 minutes.

The reason for this failure mode not presenting itself in the test was the highly local nature of the heating. In such a localised fire the epoxy at the midspan may have undergone glass transition, while epoxy on either side retained its integrity thereby keeping the tendon anchored and in tension. According to uniform compartment fire models, the most commonly used model in SFE, the beam is considered to be heated evenly along its entire length, the epoxy would accordingly de-bond along the entire length, the tendon would be released, and pre-tensioning would be lost. Furthermore, if a localised heat where to be applied at one end of a beam, although the entirety of the epoxy would not de-bond, one end of the tendon would be released leading to a loss of prestressing at the end of the beam – also potentially leading to a collapse. This logic extends to the case of a slab where the pre-tensioning plays an equally crucial role in resisting flexural stresses.

The second failure mode under consideration is longitudinal cracking observed in the fire test conducted by Sebastian and Webb. In this failure mode thermal gradients within the member section cause the stone to crack, exposing the epoxy directly to the flame which results in combustion of the epoxy – once again the assumption is made that when the integrity of the epoxy is undermined the flexural strength of the system can not be guaranteed and failure can be assumed.

In my analysis the minimum cover to insulate the strand-epoxy component and prevent debonding will be found first – for four types of common building stone. The analysis will then go on to assess thermal stresses within the stone and seek to ascertain the extent to which cracking has occurred before the debonding failure.

4. Methodology

4.1. Problem Set Up

The analysis shall consider a stone cylinder of infinite length (L) and radius (r). The stone cylinder is concentrically stressed with the strand-epoxy component occupying a hole with a radius (r_h) of 0.01m bored through the centre – dimensions consistent with the specimens fabricated by Sebastian and Webb. The amount of cover (c) therefor corresponds to the difference between the two radii ($c = r - r_h$). While the hole radius is fixed the cylinder radius acts as the primary independent variable in the analysis, adjusted stepwise by increments of 0.02m between a minimum of 0.03m and maximum of 0.41m.

Figure 17: Problem setup.

The cylinder shall be subjected to fire conditions from all sides, with a heat flux applied along its entire circumference, for a period of two hours *(t_{max} = 120 [min])* – and is as such an axisymmetric problem where heat conduction in the solid occurs along a single dimension. The second variable introduced into the analysis is stone type. The analysis seeks to compare the relative performance of four major building stone types - Limestone, Sandstone, Granite, and Marble – and ascertain which of the two failure modes under consideration – debonding of epoxy or longitudinal cracking – each stone is susceptible to.

4.2. Selection of material values

This parametric study evaluates 4 types of stone: marble, limestone, granite, sandstone. As discussed previously there is a lack of data regarding the temperature dependence of natural stones physical characteristics – for the construction of the thermal model specific heat capacity, thermal conductivity and density where required while for the mechanical model young's modulus, poisons ratio and coefficient of thermal expansion where required. While the literature has shown that all these properties are temperature dependant for all these parameters, data of sufficiently high quality was not available for all these stones. The analysis has integrated temperature dependant values for thermal conductivity, specific heat capacity and coefficient of thermal expansion. As data did not cover the full temperature range the last value was assumed and uniformly extrapolated. Characteristic values for each stone at room temperature are contained within table Table 3. And is accompanied by graphs displaying the property-temperature functions. From the literature a characteristic value for the glass transition temperature of the epoxy resin was taken to be T_g = 60C.

Table 2: Material Properties of Stone at Room Temperature[4], [65].

Property	Units	Limestone	Granite	Marble	Sandstone
Thermal Conductivity (k)	(W/m^2)	1.2950	2.855	2.505	2.365
Specific Heat Capacity (C_p)	(W.s/kG.K)	909	790	880	710
Density (ρ)	(kg/m^3)	2700	2634	2847	2341
Coefficient of Thermal Expansion (α)	(m.10^{-6}/C)	4.75	7.5	5.5	11.25
Youngs Modulus (E)	(GPa)	45	35	65	35
Poisson's Ratio (v)	-	0.270	0.230	0.276	0.250
Tensile Strength (σ_t)	(MPa)	7.00	9.83	5.78	3.02
Compressive Strength (σ_t)	(MPa)	112	150	127	44

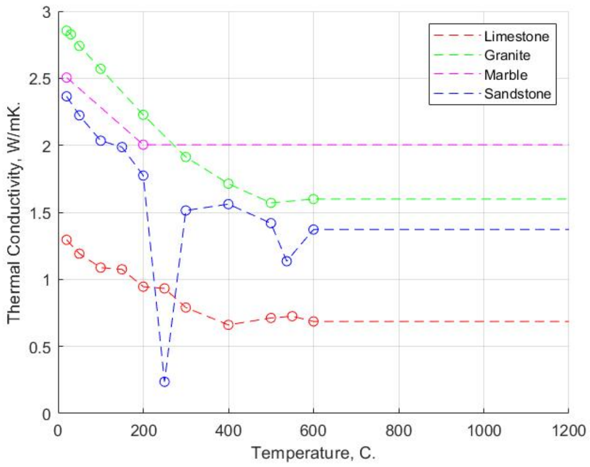

Figure 18: Thermal Conductivity as a Function of Temperature [Authors Own].

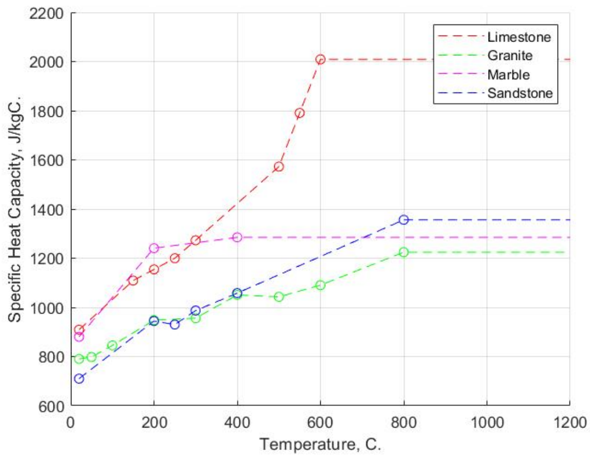

Figure 19: Specific Heat Capacity as a Function of Temperature [Authors Own].

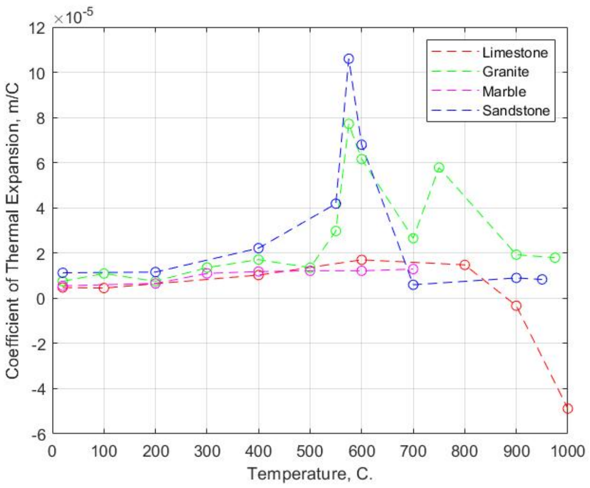

Figure 20: Coefficient of Thermal expansion as a Function of Temperature. [Authors Own]

4.3. Thermal Response Modelling

The first step in the analysis is building an accurate model of how temperatures within the element evolve over time. Assuming an infinite length means that temperature gradients within the section are consistent along the entire length of the element and allows simplification from three to two dimensions. This assumption is warranted when length is the largest dimension by a significant degree and heat effects from the ends of the element are negligible. Therefore, the thermal and subsequent mechanical analysis shall be based upon a 2D model representing the member section. The member section has been modelled as a circle with radius (r) and a hole of radius (r_c) at its centre. The omission of the epoxy-strand component is a simplification that allows the problem domain to take the form of a simple homogenous region:

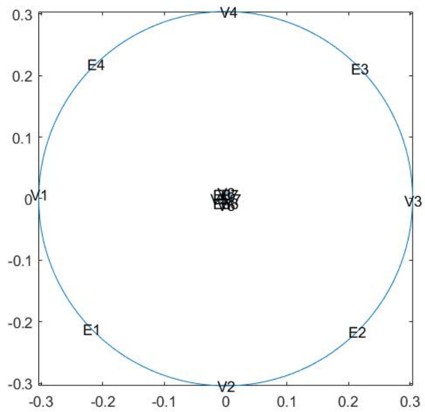

Figure 22: Example of problem domain.

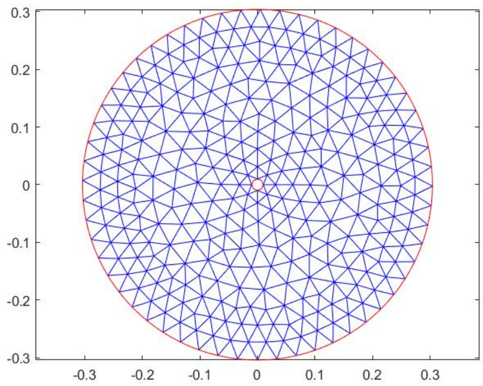

Figure 22: Example of mesh.

Due to the low thermal diffusivity of stone, and temporal nature of the question posed, the analysis demands a transient analysis. Temperatures within the stone section where therefor governed by the following heat equation:

$$\frac{\delta}{\delta x}\left(k(T)\frac{\partial T}{\partial x}\right)+\frac{\delta}{\delta y}\left(k(T)\frac{\partial T}{\partial y}\right)=\rho c_p(T)\frac{\delta T}{\delta t}$$

Where density (ρ), thermal conductivity (k) and thermal heat capacity (C_p) describe the stones thermal behaviour – the latter two of where implemented at temperature dependant functions. As well as being of a reduced dimension the heat equation assumes no internal heat generation and isotropic thermal behaviour. The governing equation is in the form of a partial differential equation, a code was built using MATLAB to execute a numerical solution based on the principles of finite element approximation. In the finite element method, the domain is first discretised giving a mesh, the PDE is evaluated at mesh nodes and then interpolated over the entire domain using shape functions. For the solution of the governing equation boundary and initial conditions, representing the problem parameters are required. The initial condition defined the initial temperature of the section at 20 degrees (T_i = 20). The internal boundary (Γ_h) is defined as adiabatic, or perfectly insulated, meaning no heat flux takes place across the surface and takes the form:

$$0 = -k\frac{dT(0,T)}{dx}$$

The consequence of omitting the epoxy-strand component and applying the of the above boundary condition means that in the model instead heat will accumulate at the interface whereas, in reality, it would pass and be stored in the epoxy. However, it is assumed that this discrepancy would be negligible due to the low volume and low thermal conductivity of the epoxy (0.15-0.25 W/mK)[66].

Along the outside edge of the cylinder (Γ_c) a heat flux density (h_{net}) was explicitly defined and applied with the following boundary condition:

$$\dot{h}_{net} = -k(T)\frac{dT(0,T)}{dx}$$

The heat flux density has two constituent parts, representing the heat gained due to convection ($\dot{h}_{net,c}$) and radiation ($\dot{h}_{net,r}$):

$$\dot{h}_{net} = \dot{h}_{net,c} + \dot{h}_{net,r}$$

Were,

$$\dot{h}_{net,c} = \alpha_c(\theta_g - \theta_m)$$

and

$$\dot{h}_{net,r} = \Phi \cdot \varepsilon_m \cdot \varepsilon_f \cdot \sigma \cdot [(\theta_g + 273)^4 - (\theta_m + 273)^4]$$

Both terms are a function of the compartment gas temperature (Θ_g) as well as several other parameters which were selected according to Eurocode suggestions. The gas temperature was based upon the standard ISO fire curve that is given by,

$$\theta_g = 20 + 345\log_{10}(8t + 1)$$

and takes the following form:

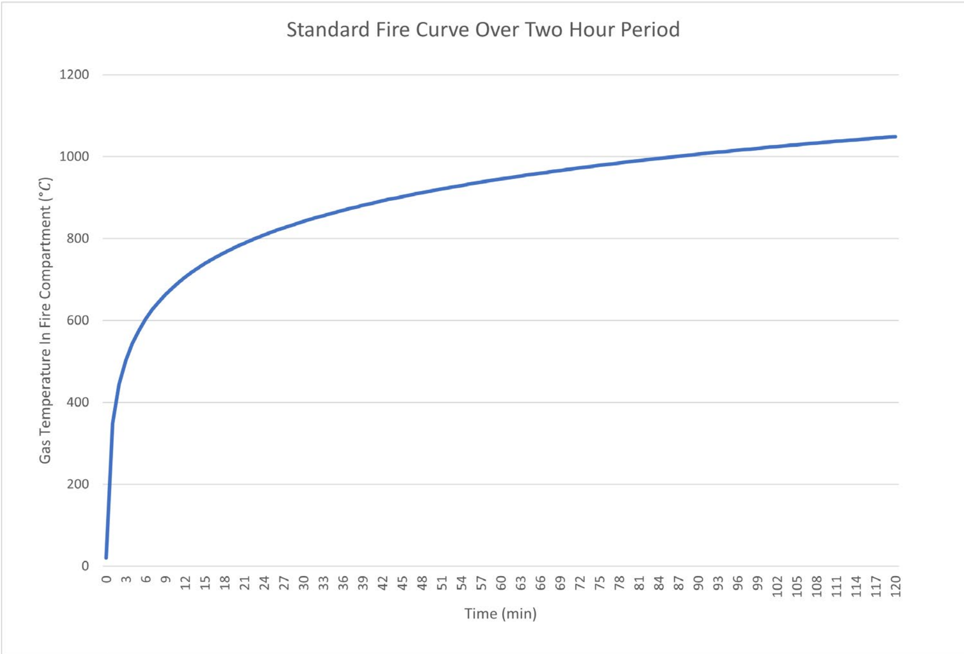

Figure 23: Compartment Gas Temperatures according to Standard ISO Fire Curve.

Probabilistic Fire Model

A probabilistic fire curve was also constructed and generated in MATLAB using parametric fire curve equations specified in the Eurocodes as a framework. This was done with the aim of giving a general idea of the amount of insulation that might be required to prevent debonding failure and achieve safety. As discussed above, the exact amount of cover is dependent on the fire, which is furthermore a function of the compartment characteristics. A deterministic evaluation for a single compartment would give limited insight. A probabilistic approach in which a multitude of fires are generated, and the analysis is realised for each case would give a broader insight into the range of minimum covers that could be required. However, due to the computational time required to analyse each case (approximately 6 hrs) this approach was not feasible and the ISO model was deemed sufficient for a comparative analysis. Details probabilistic fire model have been included in the appendix.

4.4. Mechanical Response Analysis

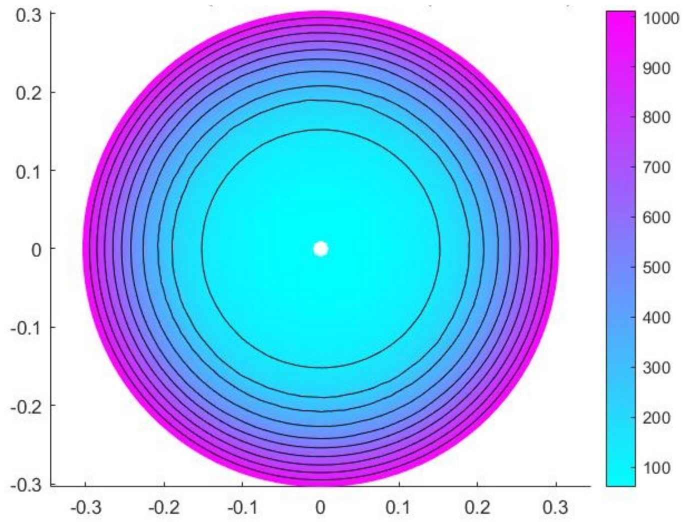

Figure 24: Example of temperature distribution obtained from MATLAB code at t = 120.

Once the thermal analysis had run the temperature profile $T_t(x)$ along the radial axis was extracted, for a given time *t*, where *x* is the radial displacement. Temperature variation $\theta_t(x)$ was then computed as the difference between the temperature profile and initial temperature: $\theta_t(x) = T_t(x) - T_{initial}$.

As the coefficient of thermal expansion α is modelled as a temperature dependant property it too was expressed as a profile, being a function of *T*, which is in turn a function of *x*: $\alpha_t(x) = f_\alpha(T_t(x))$. The function ($f_\alpha$), that relates *T* to α, takes the form of a lookup table based on the interpolated data taken from the literature.

Static plane-strain condition, where out of plane strains are set to zero, was then assumed. This assumption is representative of a slice through the middle of a long member were any out of plane expansion is resisted by the adjacent material. Radial stresses ($\sigma_{r,t}$) and hoop stresses ($\sigma_{\phi,t}$), at given time t, could subsequently be defined as a function of radial displacement (x)[46]:

$$\sigma_{r,t}(x) = \frac{E}{1-v}\left[\frac{1}{b^2 - a^2}\left(1-\frac{a^2}{x^2}\right)\int_a^b \theta_t(x)\alpha_t(x)x\,dx - \frac{1}{x^2}\int_a^x \theta_t(x)\alpha_t(x)x\,dx\right]$$

$$\sigma_{\phi,t}(x) = \frac{E}{1-v}\left[\frac{1}{b^2 - a^2}\left(1+\frac{a^2}{x^2}\right)\int_a^b \theta_t(x)\alpha_t(x)x\,dx + \frac{1}{x^2}\int_a^x \theta_t(x)\alpha_t(x)x\,dx - \theta_t(x)\alpha_t(x)\right]$$

where material properties, *Youngs modulus(E)* and *Poissons Ratio(v)*, as well as geometric properties, *inside radius (a = r_h)* and *outside radius (b = r)*, are constants.

4.5. Failure Criteria

Eurocode 0 (EN 1990) delineates the design values of actions in the case of an accidental design situation (i.e. fire) and the criteria for failure:

$$E_{d,fi} = \sum_{j\geq 1} G_{k,j} + P_k + A_k + \psi_{1/2,1}Q_{k,1} + \sum_{i>1}\psi_{2,i}Q_{k,i}$$

where $G_{k,j}$ is the characteristic value of permanent actions, $Q_{k,1}$ is the characteristic value of the main variable action, $Q_{k,i}$ subsequent variable actions, P_k the prestressing action and A_k the accidental action. Accordingly, it should be verified that for the entire duration of the fire:

$$E_{d,fi} \leq R_{fi,d}$$

where, $R_{fi,d}$ is the design resistance (capacity) in the fire scenario and $E_{d,fi}$ is the action (demand). In the case failure through debonding and longitudinal cracking it is assumed that the capacity is negligible as the epoxy is compromised and pretension lost, therefor an evaluation of the demand is not necessary.

4.5.1. Debonding of Epoxy

For the debonding of the epoxy, the performance function, by which failure can be evaluated, is expressed in terms of temperature:

$$T_{strand} \geq T_g \rightarrow R_{fi,d} = 0$$

Where, the $T_{x=rh}$ is the temperature at the strand-epoxy-stone interface and T_g is the glass transition temperature.

4.5.1. Hoop and Radial Cracking

Rankin's maximum stress theory states that brittle failure occurs when the largest principal stress reaches the materials ultimate tensile strength.

$$|\sigma_1| \geq \sigma_{ult} \text{ or } |\sigma_2| \geq \sigma_{ult}$$

Hoop and radial stresses would be plotted and compared to the material's ultimate strengths to gauge where and when failure occurred and whether in the tangential or radial direction.

5. Results and Discussion

5.1. Minimum Cover Required for Prevention of Debonding Failure

The results and discussion will begin with a comparison of the minimum cover required to prevent debonding failure for each type of stone. For each stone the analysis was run for 0.03 < r < 0.41 and the temperature at the core was recorded at the end of the run time.

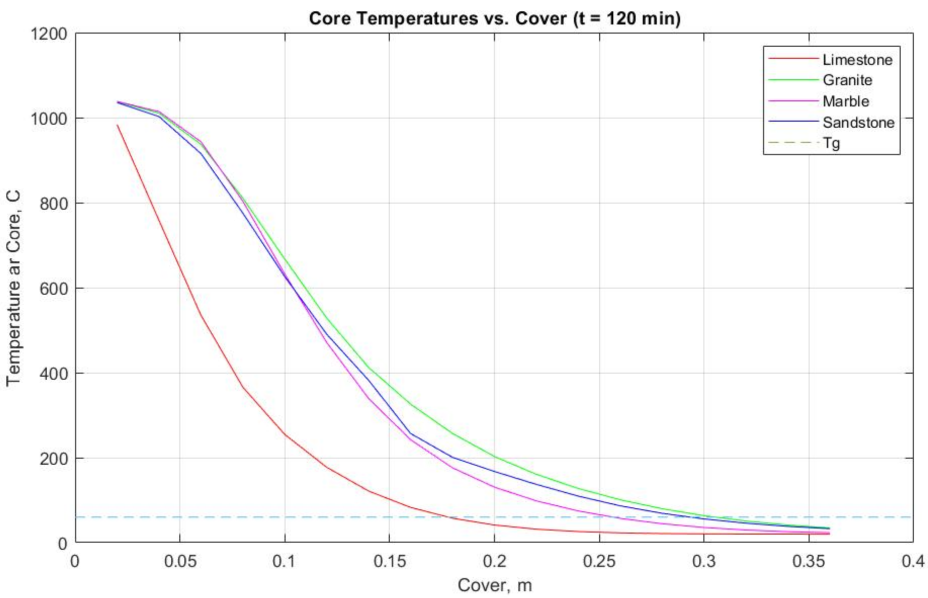

Figure 25: Core Temperature as a Function of Cover (t=120).

Figure 22. plots the results for a run time of two hours (t = 120 min) - the maximum required fire time as well as the glass transition temperature (Tg = 60 C). The temp-cover relationships for each stone show a general likeness to each other. Granite, Marble and Sandstone taking an s-shaped curve while Limestone does not display the same concavity at low covers and bears more likeness to a standard decay. The difference in shape can be understood as a function of the distinct nature of each stone's thermal conductivity, and specific heat capacity, temperature relationship which would lead to a non-linear thermal response.

From the graph one can see that limestone provides the best insulation, followed by Marble, Sandstone and then Granite. The required minimum cover corresponds to where the glass transition temperature and the recorded core temperatures intersect. The biggest disparity in the group is between limestone and marble, which require minimum covers of 0.178m and 0.257m respectively – Sandstone (0.294m) and Granite (0.305m) follow soon after. From the graph one can also see that an increased T_g value would reduce this disparity.

Table 3: Minimum Cover (m) as a function of required fire time and stone type.

Time [min]	**Limestone**	**Granite**	**Marble**	**Sandstone**
t = 120	0.178	0.305	0.257	0.294
t = 90	0.152	0.295	0.215	0.250
t = 60	0.119	0.205	0.171	0.198
t = 30	0.078	0.134	0.110	0.131

Table 4 contains the minimum cover requirements for each stone and each of the required fire resistance times that can be prescribed by the codes (30, 60, 90, 120 min). This data has been plotted in Figure 15. Which shows that the relationships discussed for $t_{fi,req}$=120 hold true for shorter times as well. As would be expected as fire time decreases the cover required decreases as does the difference between each stone. Values follow a good general trend with one exception being the performance of granite at $t_{fi,req}$=90 where there is a noticeable kink in the curve, this has again been attributed to the nonlinear response of each stone.

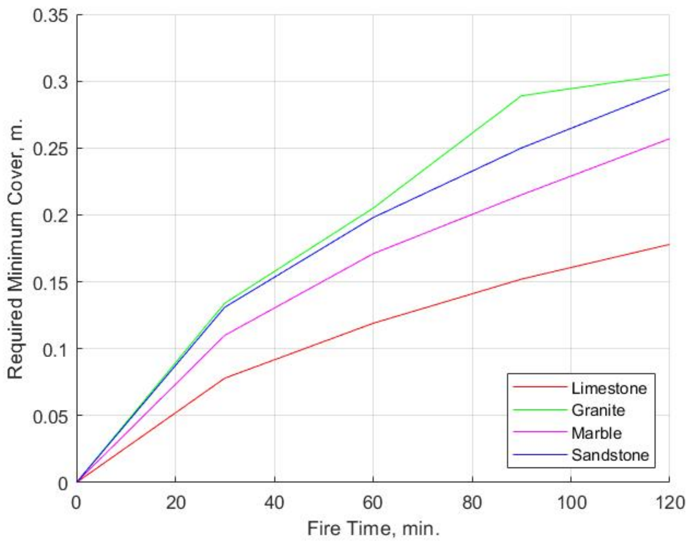

Figure 26: Required Minimum Cover vs. Fire Time.

While there is a spatial cost to a larger required minimum cover – chunkier elements take up more room, reducing ceiling height and floor space – "structural cost" is best thought of in weight. Heavier elements have a knock-on effect and necessitate a stronger and often bigger supporting structure. Heavier elements also increase transportation cost and transport emissions. The weight required per square meter of minimum cover provided has been plotted in Figure 24.

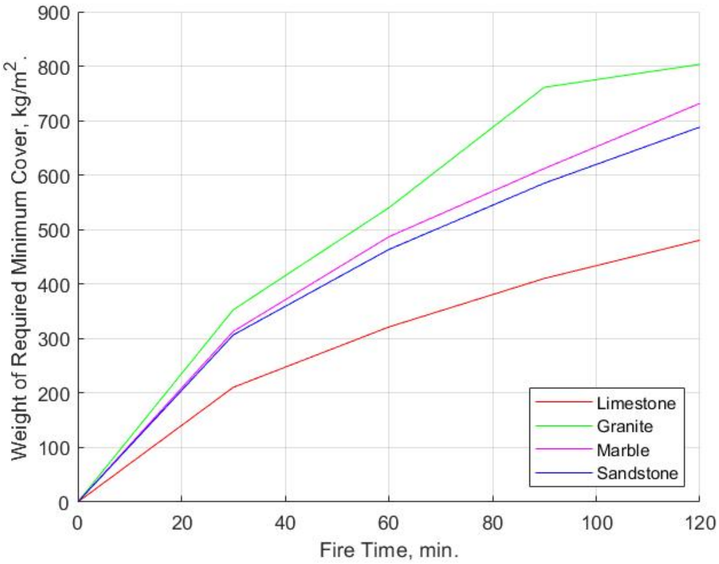

Figure 27: Weight of Required Minimum Cover as a function of Fire Time.

According to this performance criteria the ranking of the stones changes; while limestone and granite maintain their positions of best and worst respectively, in weight terms sandstone is superior to marble. This is unsurprising as of the stones under consideration, marble has the highest density (2847 kg/m^3) and sandstone, the lightest stone, has a density that is approximately 18% lower (2341 kg/m^3).

5.2. Specific Cases

The analysis will now continue to investigate four specific cases in further detail, these cases correspond to the minimum covers for protection at $t_{fi,req}$=120 for each respective stone. For each of these cases the evolution of temperatures, hoop stresses and radial stresses, over time, shall be examined.

5.2.1. Specific Case 1: Limestone (r = 0.188)

Figure 17 contains contour plots for how hoop and radial stresses in the cylinder evolve over time and distance. The characteristic strengths for tension and compression, 7MPa and -112MPa respectively are plotted as dashed lines.

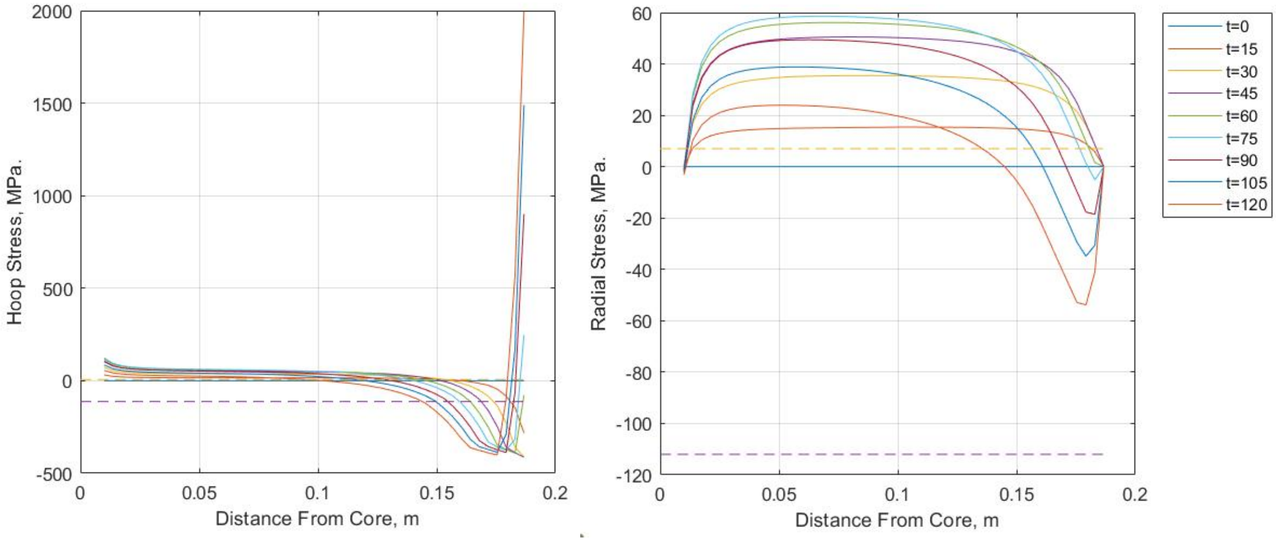

Figure 28: Limestone Specific Case, Hoop and Radial Stress Contour Plots.

Looking at the hoop stresses one can see that the both the material fails in both tension and compression almost immediately – after only 15 minutes – with the stone cracking on its interior and crushing on the exterior. At t=15 tensile strength is exceeded at a distance of 0.153m from the core and compressive strength at 0.007m from the outer surface, leaving a narrow band of the material intact. As heating continues the depth crushing proceeds at a relatively uniform rate – until a depth of 0.044m, furthermore between *60<t<75* the model computes that stresses in the stones outer surface begin to change into tension and spike dramatically.

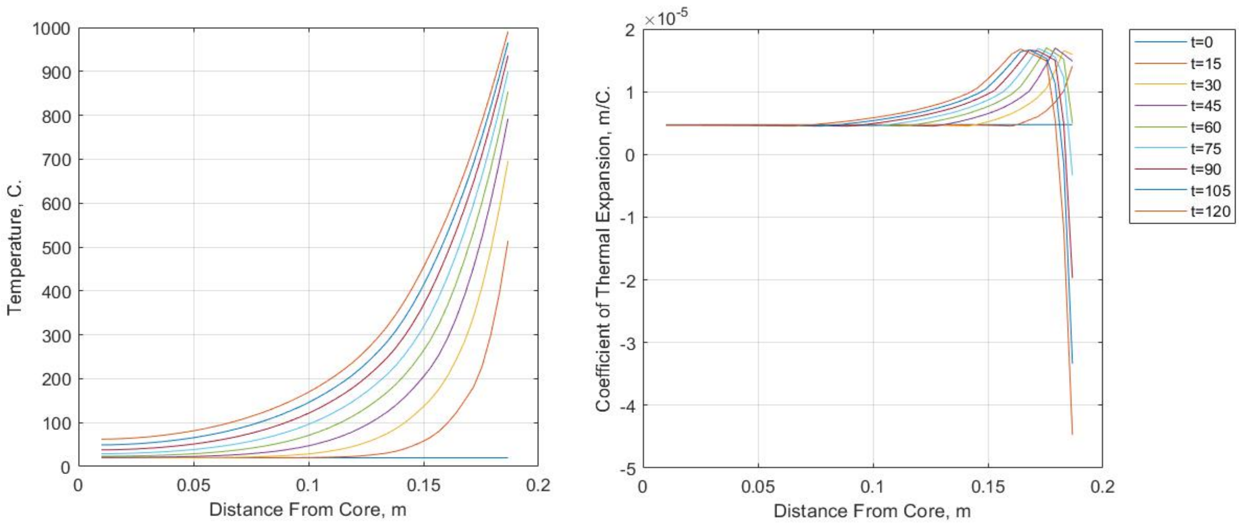

Figure 29: Limestone Specific Case, Temperature and Coefficient of Thermal Expansion Contour Plots.

Radial stresses also cause tensile failure in an equally short amount of time but are overall of a lower magnitude and never exceed the materials compressive strength. The computation shows that they remain at zero on the stones inside and outside surfaces – satisfying the expected boundary conditions. Radial stresses at the onset of heating are purely tensile, however once again between $60<t<75$ stresses at the stone's outer surface begin to switch direction and change into compression.

This distinctive change in shape can be explained by Figure 18. Which contains temperature contours and thermal expansion contours. At $t=60$ the coefficient of thermal expansion begins to decrease, by $t=75$ it is negative, and it continues to drop and is equal to -4*(10^-5) m/C by t=120. This means that the material begins to contract as opposed to expand – hence a reversal in the stresses. Looking at the corresponding temperatures these changes begin when 800C is reached – from the literature review we know that this is the temperature at which the calcium carbonate in the limestone begins to decompose.

It is worth noting that stresses recorded beyond the materials stress limits have little meaning as the material could not sustain them, and therefor they could not occur. Furthermore, once material has failed stresses would have to redistribute themselves. The computation does not take this nonlinearity into account and as stress limits are exceeded very rapidly the relevance of the magnitude of the peak stresses is questioned. A physical interpretation of the results would be that tensile stresses result in radial cracking at the core and circumferential cracking throughout, compressive stresses cause spalling on the fire exposed surface and after an hour thermal degradation on the fire exposed surface is accelerated as the calcium carbonate decomposes.

5.2.2. Specific Case 2: Marble (r = 0.267)

The characteristic strengths plotted with hoop and radial stresses in Figure 22. are 5.78MPa and -127MPa for tension and compression respectively.

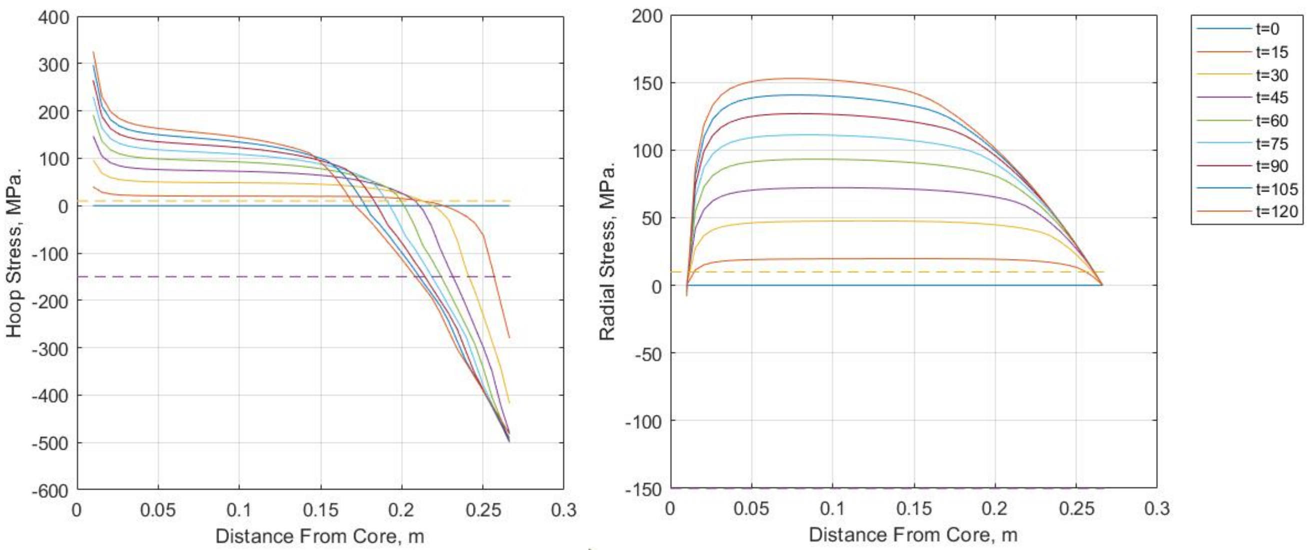

Figure 30: Marble Specific Case, Hoop and Radial Stress Contour Plots.

Looking at the hoop and radial stresses in Figure 22. one can see that the shape of stresses in marble quite different as the phenomenon of stress reversal at the end of run time is not present. Tensile and compressive stresses are still exceeded in the same time span – after just 15 minutes. Hoop stresses exceed tensile strength 0.212m from the core and compressive strength 0.01m from the outside surface. There is also a more pronounced upwards concavity where hoop stresses approach the core and stresses "flick" upwards – with the magnitude of these tensile forces three times greater than in the limestone sample. Again, these stresses could not occur in reality.

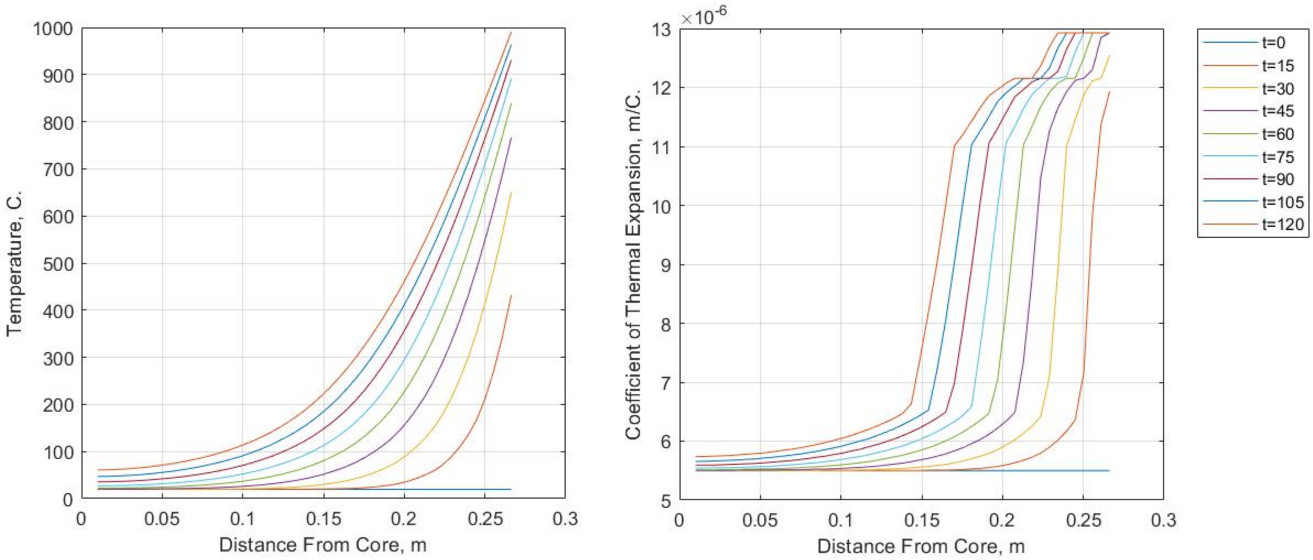

Figure 31: Marble Specific Case, Temperature and Coefficient of Thermal Expansion Contour Plots.

Figure 21. Shows a that the temperature dependence of the coefficient of thermal expansion in marble is different than in limestone, and changes in a stepwise but relatively smooth manor – thermal expansion first increases gradually until it reaches a point where there is a steep upwards branch, it then plateaus before increasing rapidly again, but by a small amount, and then levelling off. This stepwise shape could however be an artifact of the data which has been linearly interpolated. The main upwards branch occurs when T = 400. The magnitude and temperature at which this increase occurs is very similar to the initial increase observed with limestone. Looking at the graph of thermal expansion vs. temperature presented in the methodology (Figure 13.) it is notable that data for the thermal expansion cuts for marble cuts off at 700 degrees. Therefore, it is unknown whether the marble, which is known to degrade to the point of friability at extreme temperatures, would follow a similar shape to limestone and exhibit the same reversal of stresses.

5.2.3. Specific Case 3: Granite (r = 0.315)

The characteristic strengths plotted with hoop and radial stresses in Figure 19. are 9.83MPa and -150MPa for tension and compression respectively.

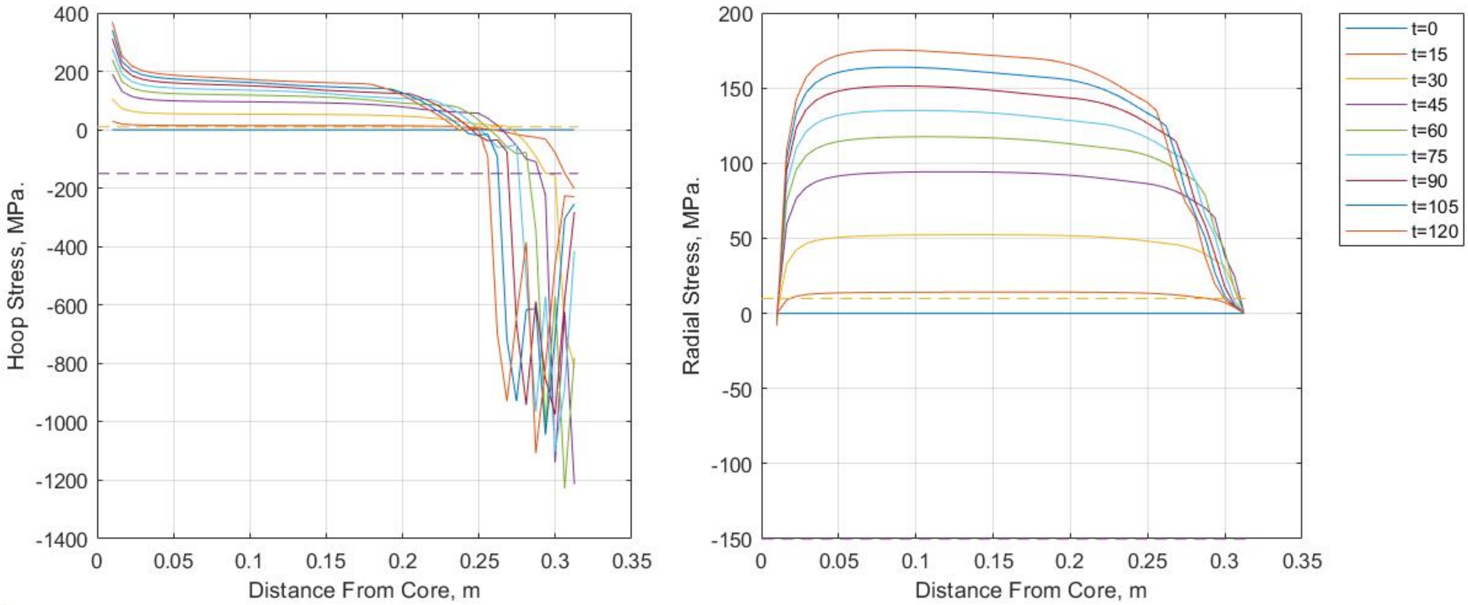

Figure 32: Granite Specific Case, Hoop and Radial Stress Contour Plots.

Plots of hoop and radial stresses show a shape distinct from both Limestone and Marble. Once again, radial cracking occurs after just 15 minutes on the interior while circumferential cracking occurs throughout the entire section and compressive stresses on the heated surface cause crushing starting at 15 minutes.

The key difference from the other plots is the dramatic jump in hoop stresses at the outer surface after 30 minutes and the volatile manor in which they oscillate from that point onwards. As stated previously, stresses recorded beyond the materials yield strength have limited meaning. The shape of the radial stresses plotted are consistent with those seen in marble.

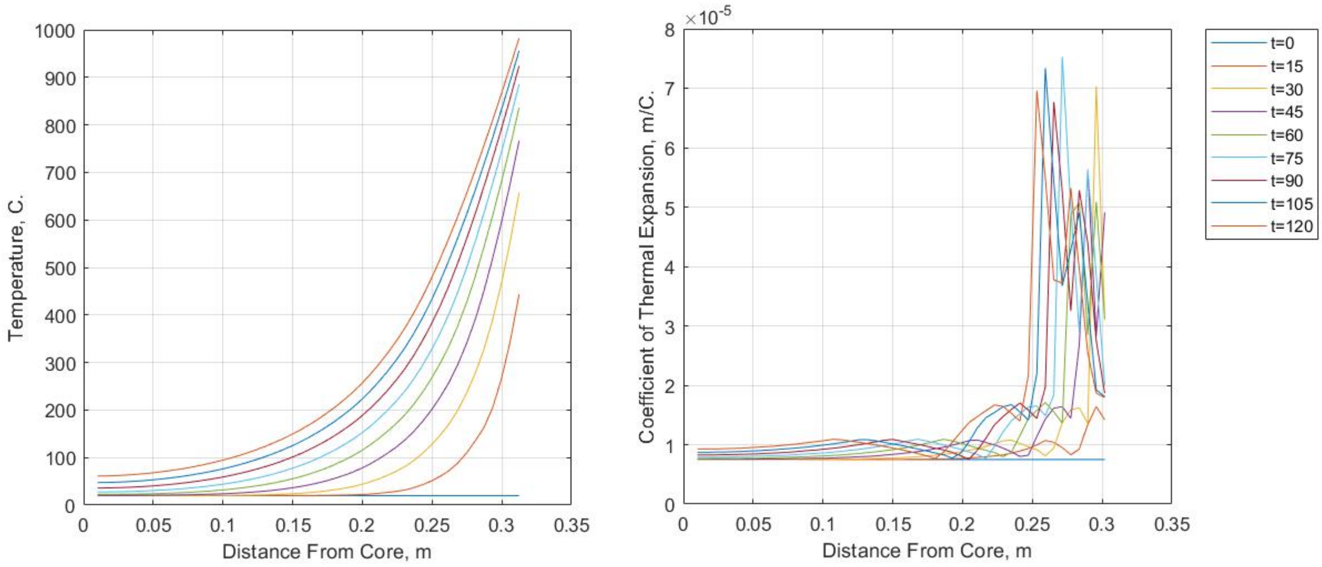

Figure 33: Granite Specific Case, Temperature and Coefficient of Thermal Expansion Contour Plots.

Figure 20. Shows us a correspondence between the patterns seen in the hoop stresses, temperature and the coefficient of thermal expansion. Relative to limestone the temperature gradient across the section is less steep, however thermal expansion peaks at a value about 4 times higher and peaks multiple times suggesting multiple cracking phases. At the time of the first peak, occurring between 15- and 30-minutes, temperatures increase from 450 to 650 degrees. This once again corresponds well with the qualitative description of stone covered in the literature review – as it is at 573 degrees that quartz minerals undergo a change in crystalline structure known to put granite at risk of shattering.

5.2.4. Specific Case 4: Sandstone (r = 0.304)

The characteristic strengths plotted with hoop and radial stresses in Figure 24. are 3.02MPa and -44MPa for tension and compression respectively.

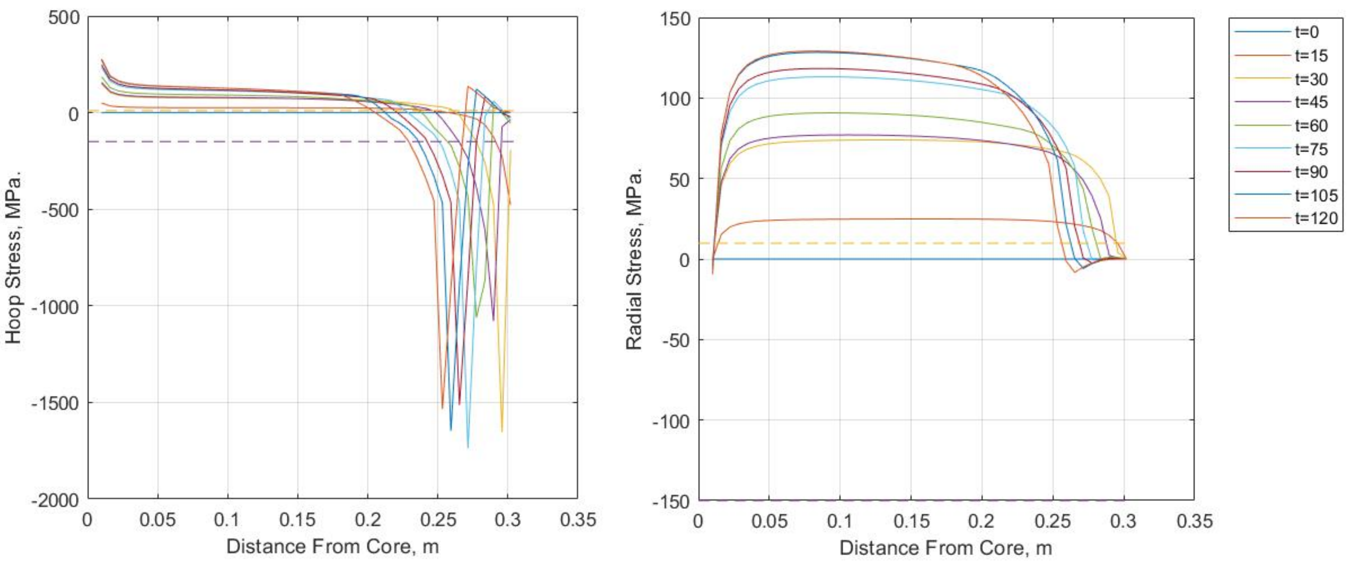

Figure 34: Sandstone Specific Case, Hoop and Radial Stress Contour Plots.

Sandstone fails within the same timeframe as the other stones. Figure 24. shows behavior that hoop and radial stresses follow a similar pattern to in granite – but with less oscillation. Instead of spiking twice stress only spikes once. Another significant difference is that, at the fire exposed surface, after the compression spikes the material goes into tension for a short period befor returning compression again. The coefficient of thermal expansion is never negative like in limestone so this is a notable obeservation. Likewise towards the end of the runtime at the outer surface of the stone radial stresses are compressive, something not seen in either granite or marble.

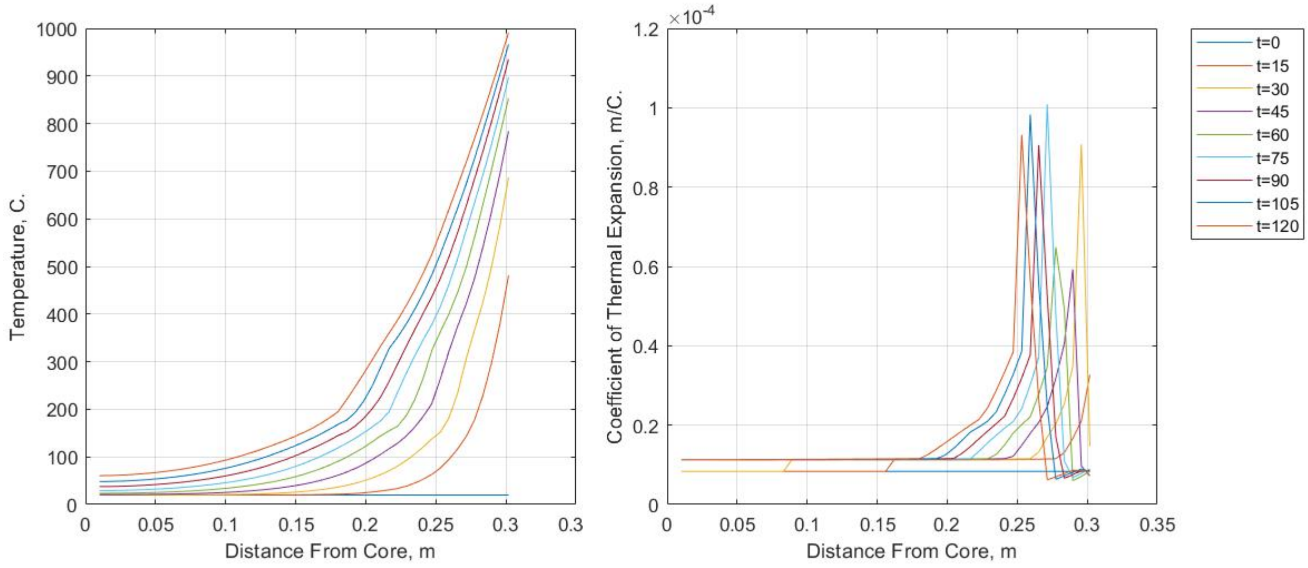

Figure 35: Sandstone Specific Case, Temperature and Coefficient of Thermal Expansion Contour Plots.

Looking at Figure 23. One can see that the spikes in compressive hoop stresses are consistent with the coefficient of thermal expansion. What is interesting is the inconsistency in magnitude of these spikes – the peek expansion at t=30 us much greater than at t=60. This non-linearity is also found in the temperature contour graph, which although following the general shape of a decay, has a less smooth temperature profile than other stones and exhibits slight but noticeable discontinuities that occur around T=200. This corresponds to a sharp but transient drop that occurs in the thermal conductivity of sandstone at the same point (Figure 12.)

5.2.5. Comparison

Figure 27 shows the rate distance between the point at which hoop stresses exceed the materials compressive strength and the outer surface in absolute terms and relative to the cylinder diameter. If the physical interpretation of these compressive stresses is spalling than sandstone is the worst affected – which can be attributed to its low compressive strength – while limestone performs the best. Marble and Granite spall at similar rates. Looking at how deep spalling penetrates relative to the cylinders radius granite performs best at 20% and sandstone worst at 25%. The overall spread is less with this adjustment.

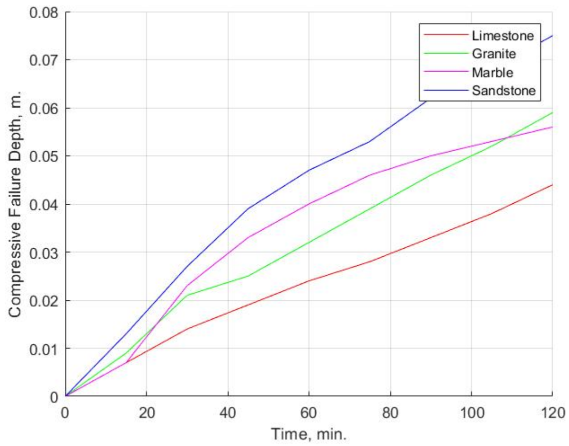

Figure 37: Depth of Compressive Failure over Time.

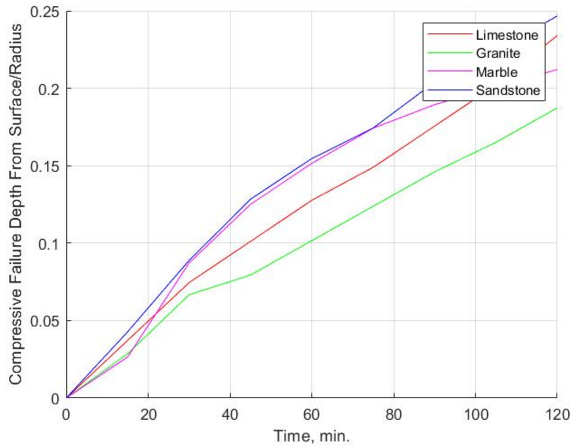

Figure 37: Relative depth of Compressive Failure over Time.

Figure 28. and Figure 29. Show the crack penetration depth after 15 minutes in both relative and absolute terms. When adjusted for the scale of the cylinder all stones come out roughly the same with the tensile strength exceeded in 75-80% of the material. In absolute terms limestone performs best followed by marble, granite and then sandstone.

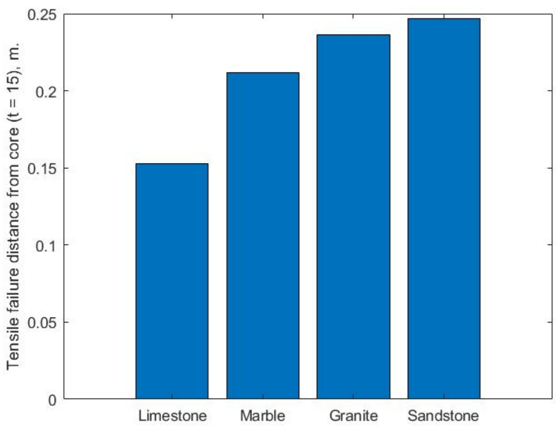

Figure 39: Depth of initial tensile failure.

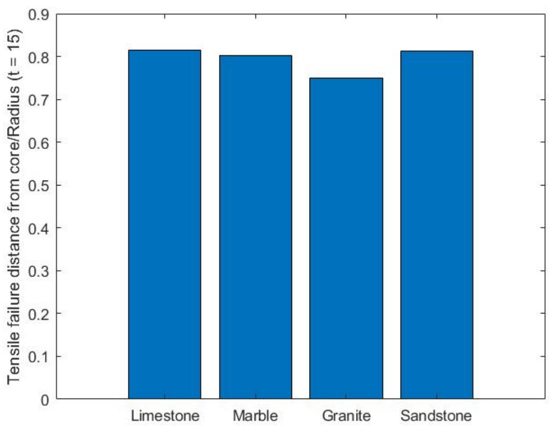

Figure 38: Depth of relative tensile failure.

6. Conclusions and Limitations

The approach used in this paper was a combination of numerical FEA model for calculating the evolution of temperature over time and space, an analytical expression for converting those temperatures into stresses and temperature dependant functions to capture the evolution of material properties as they heat.

The thermal analysis gave believable and useful results for the characteristic minimum covers one could expect for each stone type and temperature profiles within the material. The effect of the temperature dependant functions could be seen in the temperature profiles of sandstone especially. A limitation of the analysis was the lack data relating stone density to temperature – which if available could have altered the results. The comparison between each stones performance in terms of weight required for the provision of minimum cover was valuable as the difference between stone types was not insubstantial –with the weight requirement for granite being twice that of limestone. The analysis could be enriched by giving upper and lower boundary values for each stone type and a sensitivity analysis to ascertain how perturbations in material parameters would propagate through the FEA code and effect performance. The analysis could also be criticised for using an overly conservative fire model – the ISO fire curve – however as compartment characteristics where unknown, and analysis was comparative, the use of the ISO curve is warranted so long as the reader bears in mind that values are on the conservative side. Overall, it can be said with some confidence that limestone provides the best insulation with the lowest minimum cover requirement – followed by marble, sandstone and then granite.

The mechanical analysis showed a good correlation to the qualitative understanding of each stones behaviour as patterns in stresses corresponded well with the morphological changes that would occur in the stones at the corresponding temperatures. The analysis also revealed the importance of including a temperature dependant function when defining the coefficient of thermal expansion. The results also showed a consistent and sensible pattern of cracking and spalling – with tensile hoop stresses occurring on the member interior and compressive stresses on the fire exposed surface. The depth to which compressive stresses penetrated proceed at a uniform rate across the entire run time. Another finding was the presence of cracks due to radial stresses on the member interior. However, the speed at which failure occurred, after just 15 minutes, was surprising when put in context of the fire test conducted by Sebastian and Webb – in which longitudinal cracks occurred after 86 minutes. However, the fire temperatures in the analysis where far higher than those produced in the practical test. The magnitude of the stresses recorded also had limited meaning as the material could not have sustained those stresses.

The simplicity of the analytical expression used to calculate stresses limited the mechanical analysis and when interpreting the results, it is not clear at which point the epoxy could said to be exposed to the flame by longitudinal cracking. For a comprehensive understanding of crack initiation and propagation a fracture mechanics-based approach is needed that considers the plastic non-linear nature of cracking. Phenomena such as strain softening would have enriched the analysis - strain softening models allow material properties to soften or harden after the onset of plastic yield according to a certain function. Implementation of a temperature dependant function for the young's modulus – which we from the literature review we know decreases with temperature - could have captured this to some effect and led to more realistic stresses.

Furthermore, the complexity of the rocks heterogeneous mineral-void matrix would produce more accurate results. Instead of homogenous material values a probabilistic property distribution could have been implemented in the FEA code – where instead of every element having the same value, the value varies from element to element according to a suitable distribution. Some studies in the literature that have successfully simulated the cracking of stone have used 3D FEA elements to model the stone as in terms of its constitutive minerals – specific material properties for each mineral are then applied to the elements with a degree of natural variation [67]. However, such a high-fidelity investigation was beyond the scope of this paper.

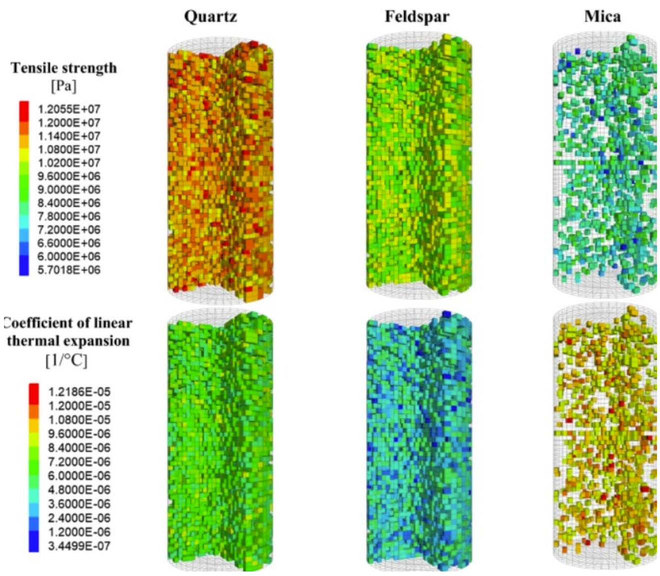

Figure 40: 3D model used to simulate thermal cracking of stone[67].

Further research could be directed towards defining a technique for the modelling of thermal cracking in stone that produces high fidelity results and can be easily implemented quickly in practice. However, from the literature review, it is evident that first gaps in data for the how physical characteristics relate to temperature must be filled by practical research. Practical research is also required regarding the glass transition of epoxy and the curing parameters that influence it. These are the two main areas of research necessary for ensuring a strand-epoxy-stone prestressing system will perform safely in the case of a fire.

7. List of Figures

8. References

[1] A. T. Balasbaneh, A. Kadir, B. Marsono, and A. Phd, “New Residential Construction Building and Composite Post and Beam Structure toward Global Warming Mitigation,” vol. 00, pp. 0–000, 2017, doi: 10.1002/ep.12807.

[2] R. Jing, M. Wahab, J. Qian, and Z. Zhang, “Assessments of Greenhouse Gas (GHG) Emissions from Stainless Steel Production in China Using Two Evaluation Approaches,” 2019, doi: 10.1002/ep.13125.

[3] S. Oyebisi, A. Ede, F. Olutoge, and D. Omole, “Geopolymer concrete incorporating agro-industrial wastes: Effects on mechanical properties, microstructural behaviour and mineralogical phases,” *Construction and Building Materials*, vol. 256, p. 119390, Sep. 2020, doi: 10.1016/J.CONBUILDMAT.2020.119390.

[4] B. Hunt, “Building stones explained 4,” *Geology Today*, vol. 22, pp. 33–40, Aug. 2006, doi: 10.1111/j.1365-2451.2006.00547.x.

[5] “Natural Stone the oldest sustainable material.”

[6] W. Yates Engineers, S. BOOTE BEng, and A. Associate, “Stone as a structural material. Part 1: Mechanical properties.”

[7] “https://blog.polycor.com/a-return-to-form-the-renaissance-of-structural-stone.”

[8] Heyman, *The stone skeleton*. 1966.

[9] W. Yates Engineers, “Stone-as-a-structural-material-Part-3-Post-tensioned-stone-structures”.

[10] W. Sebastian and S. Webb, “Tests on prototype pretensioned natural stone beams,” *Construction and Building Materials*, vol. 271, p. 121555, Feb. 2021, doi: 10.1016/J.CONBUILDMAT.2020.121555.

[11] “https://www.concretenetwork.com/post-tension/.”

[12] W. Yates Engineers, S. BOOTE BEng, and A. Associate, “Stone as a structural material. Part 2: Traditional and reinforced stone stairs,” 2020.

[13] W. Yates Engineers *et al.*, “Stone as a structural material. Part 4: Contemporary loadbearing stone buildings.”

[14] Kuhn. S, “Prestressed Stone Panels: La Sagrada Familia. ,” ETHZ , 2022.

[15] M. Darot and T. Reuschlé, “Acoustic wave velocity and permeability evolution during pressure cycles on a thermally cracked granite,” *International Journal of Rock Mechanics and Mining Sciences*, vol. 37, no. 7, pp. 1019–1026, Oct. 2000, doi: 10.1016/S1365-1609(00)00034-4.

[16] H. Yavuz, S. Demirdag, and S. Caran, “Thermal effect on the physical properties of carbonate rocks,” *International Journal of Rock Mechanics and Mining Sciences*, vol. 47, no. 1, pp. 94–103, Jan. 2010, doi: 10.1016/J.IJRMMS.2009.09.014.

[17] Hajpál, “Fire Damaged Stone Structures in Historical Monuments . Laboratory Analyses of Changes in Natural Stones by Heat Effect,” 2010.

[18] B. Chakrabarti, T. Yates, and A. Lewry, “Effect of fire damage on natural stonework in buildings,” *Construction and Building Materials*, vol. 10, no. 7, pp. 539–544, 1996, doi: 10.1016/0950-0618(95)00076-3.

[19] A. Biró, V. Hlavička, and É. Lublóy, “Effect of fire-related temperatures on natural stones,” *Construction and Building Materials*, vol. 212, pp. 92–101, Jul. 2019, doi: 10.1016/J.CONBUILDMAT.2019.03.333.

[20] A. Biró, V. Hlavička, and É. Lublóy, “Effect of fire-related temperatures on natural stones,” *Construction and Building Materials*, vol. 212, pp. 92–101, Jul. 2019, doi: 10.1016/J.CONBUILDMAT.2019.03.333.

[21] E. Martinho and A. Dionísio, “Assessment Techniques for Studying the Effects of Fire on Stone Materials: A Literature Review,” *International Journal of Architectural Heritage*, vol. 14, pp. 1–25, Jul. 2018, doi: 10.1080/15583058.2018.1535008.

[22] S. Sci, L. Zhang, M. Xianbiao, and L. U. Aihong, “SCIENCE IN CHINA PRESS Experimental study on the mechanical properties of rocks at high temperature,” *China Ser E-Tech Sci*, vol. 52, no. 3, pp. 641–646, 2009, doi: 10.1007/s11431-009-0063-y.

[23] E. Franzoni, E. Sassoni, G. W. Scherer, and S. Naidu, “Artificial weathering of stone by heating,” *Journal of Cultural Heritage*, vol. 14, no. 3, pp. e85–e93, Jun. 2013, doi: 10.1016/J.CULHER.2012.11.026.

[24] A. Ozguven and Y. Ozcelik, “Effects of high temperature on physico-mechanical properties of Turkish natural building stones,” *Engineering Geology*, vol. 183, pp. 127–136, Dec. 2014, doi: 10.1016/J.ENGGEO.2014.10.006.

[25] B. Chakrabarti, T. Yates, and A. Lewry, “Effect of fire damage on natural stonework in buildings,” *Construction and Building Materials*, vol. 10, no. 7, pp. 539–544, Oct. 1996, doi: 10.1016/0950-0618(95)00076-3.

[26] W. G. Liang, S. G. Xu, and Y. S. Zhao, “Experimental Study of Temperature Effects on Physical and Mechanical Characteristics of Salt Rock,” *Rock Mech. Rock Engng*, no. 5, pp. 469–482, 2006, doi: 10.1007/s00603-005-0067-2.

[27] A. Ozguven and Y. Ozcelik, “Investigation of some property changes of natural building stones exposed to fire and high heat,” *Construction and Building Materials*, vol. 38, pp. 813–821, Jan. 2013, doi: 10.1016/J.CONBUILDMAT.2012.09.072.

[28] F. Sciarretta, J. Eslami, A.-L. Beaucour, and A. Noumowé, “State-of-the-art of construction stones for masonry exposed to high temperatures,” *Construction and Building Materials*, vol. 304, p. 124536, 2021, doi: https://doi.org/10.1016/j.conbuildmat.2021.124536.

[29] D. M. Freire-Lista, R. Fort, and M. J. Varas-Muriel, “Thermal stress-induced microcracking in building granite,” *Engineering Geology*, vol. 206, pp. 83–93, May 2016, doi: 10.1016/J.ENGGEO.2016.03.005.

[30] Z.-H. Wang, H.-W. Zhou, H. Konietzky, S.-M. Li, T.-Z. Wang, and T. Wang, “Physical and mechanical properties of Beishan granite after different high temperature treatments,” *J. Cent. South Univ*, vol. 2022, no. 6, pp. 1957–1972, 2022, doi: 10.1007/s11771-022-5063-0.

[31] D. M. Freire-Lista, R. Fort, and M. J. Varas-Muriel, “Thermal stress-induced microcracking in building granite,” *Engineering Geology*, vol. 206, pp. 83–93, May 2016, doi: 10.1016/J.ENGGEO.2016.03.005.

[32] F. Wang and H. Konietzky, “Thermo-Mechanical Properties of Granite at Elevated Temperatures and Numerical Simulation of Thermal Cracking,” *Rock Mechanics and Rock Engineering*, vol. 52, pp. 3737–3755, 2019, doi: 10.1007/s00603-019-01837-1.

[33] M. Hajpál, “Changes in Sandstones of Historical Monuments Exposed to Fire or High Temperature,” 2002.

[34] J. Michels, R. Widmann, C. Czaderski, R. Allahvirdizadeh, and M. Motavalli, “Glass transition evaluation of commercially available epoxy resins used for civil engineering applications,” *Composites Part B: Engineering*, vol. 77, pp. 484–493, 2015, doi: https://doi.org/10.1016/j.compositesb.2015.03.053.

[35] P. Kumar, A. Patnaik, and S. Chaudhary, “A review on application of structural adhesives in concrete and steel–concrete composite and factors influencing the performance of composite connections,” *International Journal of Adhesion and Adhesives*, vol. 77, pp. 1–14, Sep. 2017, doi: 10.1016/J.IJADHADH.2017.03.009.

[36] B. F. J. F. G, T. A. J. B, and N. J. A. D, “Bond Geometry and Shear Strength of Steel Plates Bonded to Concrete on Heating,” *Journal of Materials in Civil Engineering*, vol. 15, no. 6, pp. 586–593, Dec. 2003, doi: 10.1061/(ASCE)0899-1561(2003)15:6(586).

[37] L. C. and L. M. Hollaway, *Strengthening of reinforced concrete structures: Using externally-bonded FRP composites in structural and civil engineering*. Elsevier, 1999.

[38] H. R. and C. R. T. and others Sasse, *Adhesion between polymers and concrete*. Springer, 2013.

[39] E. U. Chowdhury, L. A. Bisby, M. F. Green, and V. K. R. Kodur, “Investigation of insulated FRP-wrapped reinforced concrete columns in fire,” *Fire Safety Journal*, vol. 42, no. 6–7, pp. 452–460, Sep. 2007, doi: 10.1016/J.FIRESAF.2006.10.007.

[40] N. Pinoteau, P. Pimienta, T. Guillet, P. Rivillon, and S. Rémond, “Effect of heating rate on bond failure of rebars into concrete using polymer adhesives to simulate exposure to fire,” *International Journal of Adhesion and Adhesives*, vol. 31, no. 8, pp. 851–861, Dec. 2011, doi: 10.1016/J.IJADHADH.2011.08.005.

[41] S. Khalaf Mohaisen, “EXPERIMENTAL BOND FORCE-SLIP RELATIONSHIPS FOR EPOXY-COATED REINFORCING BARS UNDER ELEVATED TEMPERATURE CONDITIONS,” *Printed in Iraq*, vol. 05, no. 01, pp. 160–171, 2012.

[42] N. Pinoteau, P. Pimienta, T. Guillet, P. Rivillon, and S. Rémond, “Effect of heating rate on bond failure of rebars into concrete using polymer adhesives to simulate exposure to fire,” *International Journal of Adhesion and Adhesives*, vol. 31, no. 8, pp. 851–861, Dec. 2011, doi: 10.1016/J.IJADHADH.2011.08.005.

[43] R. Kotynia, K. Adamczewska, A. Strakowska, M. Masłowski, and K. Strzelec, “Effect of Accelerated Curing Conditions on Shear Strength and Glass Transition Temperature of Epoxy

Adhesives,” *Procedia Engineering*, vol. 193, pp. 423–430, Jan. 2017, doi: 10.1016/J.PROENG.2017.06.233.

[44] B. Williams, V. Kodur, M. F. Green, and L. Bisby, “Fire Endurance of Fiber-Reinforced Polymer Strengthened Concrete T-Beams,” *ACI Structural Journal*, vol. 105, no. 1, pp. 60–67, 2008, [Online]. Available: https://www.proquest.com/scholarly-journals/fire-endurance-fiber-reinforced-polymer/docview/198339744/se-2?accountid=14511

[45] A. Usmani, J. Rotter, S. Lamont, A. M. Sanad, and M. Gillie, “Fundamental principles of structural behaviour under thermal effects,” *Fire Safety Journal*, vol. 36, pp. 721–744, Jul. 2001, doi: 10.1016/S0379-7112(01)00037-6.

[46] H. Richard B and E. M. Reza, *Thermal Stresses -- Advanced Theory and Applications*. 2009.

[47] V. Saseendran, “Fracture Characterization and Analysis of Debonded Sandwich Composites,” 2017. doi: 10.13140/RG.2.2.18534.98882.

[48] J. A. Purkiss, *Fire safety engineering : design of structures / John A. Purkiss.*, 2nd ed. Oxford: Butterworth-Heinemann, 2006.

[49] Great Britain. Department for Communities and Local Government., *Approved document B. Fire safety.* NBS/RIBA Publishing, 2007.

[50] H. Francisco and B. Xavier, “ANALYSIS OF REINFORCED CONCRETE FRAMES EXPOSED TO FIRE Based on Advanced Calculation Methods,” 2009. [Online]. Available: http://www.fe.up.pt

[51] T. Gernay, R. van Coile, N. Elhami Khorasani, and D. Hopkin, “Efficient uncertainty quantification method applied to structural fire engineering computations,” *Engineering Structures*, vol. 183, pp. 1–17, Mar. 2019, doi: 10.1016/J.ENGSTRUCT.2019.01.002.

[52] “EN 1992-1-2: Eurocode 2: Design of concrete structures - Part 1-2: General rules - Structural fire design,” 1992.

[53] J. L. Torero, A. Law, and C. Maluk, “Defining the thermal boundary condition for protective structures in fire,” *Engineering Structures*, vol. 149, pp. 104–112, Oct. 2017, doi: 10.1016/J.ENGSTRUCT.2016.11.015.

[54] A. Buchanan, “The challenges of predicting structural performance in fires,” in *Fire Safety Science*, 2008, pp. 79–90. doi: 10.3801/IAFSS.FSS.9-79.

[55] A. A. Khan, A. Usmani, and J. L. Torero, “Evolution of fire models for estimating structural fire-resistance,” *Fire Safety Journal*, vol. 124, p. 103367, Sep. 2021, doi: 10.1016/J.FIRESAF.2021.103367.

[56] G. Rein *et al.*, “Round-robin study of a priori modelling predictions of the Dalmarnock Fire Test One,” *Fire Safety Journal*, vol. 44, no. 4, pp. 590–602, May 2009, doi: 10.1016/J.FIRESAF.2008.12.008.

[57] J. Stern-Gottfried and G. Rein, “Travelling fires for structural design-Part II: Design methodology,” *Fire Safety Journal*, vol. 54, pp. 96–112, Nov. 2012, doi: 10.1016/J.FIRESAF.2012.06.011.

[58] C. D. Eamon and E. Jensen, “Reliability analysis of reinforced concrete columns exposed to fire,” *Fire Safety Journal*, vol. 62, no. PART C, pp. 221–229, Nov. 2013, doi: 10.1016/J.FIRESAF.2013.10.002.

[59] C. D. Eamon and E. Jensen, “Reliability analysis of prestressed concrete beams exposed to fire,” *Engineering Structures*, vol. 43, pp. 69–77, Oct. 2012, doi: 10.1016/J.ENGSTRUCT.2012.05.016.

[60] R. van Coile, R. Caspeele, and L. Taerwe, “Reliability-based evaluation of the inherent safety presumptions in common fire safety design,” *Engineering Structures*, vol. 77, pp. 181–192, Oct. 2014, doi: 10.1016/J.ENGSTRUCT.2014.06.007.

[61] Q. Guo, K. Shi, Z. Jia, and A. E. Jeffers, “Probabilistic Evaluation of Structural Fire Resistance”, doi: 10.1007/s10694-012-0293-6.

[62] R. and C. R. and T. L. van Coile, “The mixed lognormal distribution for a more precise assessment of the reliability of concrete slabs exposed to fire”.

[63] F. Bamer, D. Thaler, M. Stoffel, and B. Markert, “A Monte Carlo Simulation Approach in Non-linear Structural Dynamics Using Convolutional Neural Networks,” *Frontiers in Built Environment | www.frontiersin.org*, vol. 1, p. 679488, 2021, doi: 10.3389/fbuil.2021.679488.

[64] K. Olsson, J. Anderson, and D. Lange, “Uncertainty propagation in FE modelling of a fire resistance test using fractional factorial design based model reduction and deterministic sampling,” *Fire Safety Journal*, vol. 91, pp. 517–523, Jul. 2017, doi: 10.1016/J.FIRESAF.2017.03.032.

[65] F. Sciarretta, J. Eslami, A.-L. Beaucour, and A. Noumowé, “State-of-the-art of construction stones for masonry exposed to high temperatures,” *Construction and Building Materials*, vol. 304, p. 124536, 2021, doi: https://doi.org/10.1016/j.conbuildmat.2021.124536.

[66] S. L. Chung and J. S. Lin, “Thermal conductivity of epoxy resin composites filled with combustion synthesized h-BN particles,” *Molecules*, vol. 21, no. 5, May 2016, doi: 10.3390/molecules21050670.

[67] F. Wang, H. Konietzky, T. Frühwirt, and Y. Dai, “Laboratory testing and numerical simulation of properties and thermal-induced cracking of Eibenstock granite at elevated temperatures,” *Acta Geotechnica*, vol. 15, no. 8, pp. 2259–2275, 2020, doi: 10.1007/s11440-020-00926-8.